CONDITIONING A MEASURE-VALUED DIFFUSION ON THE WASSERSTEIN SPACE

FIRST LAST

ABSTRACT. A concise abstract summarizing the problem, main results, and significance.

CONTENTS

1. INTRODUCTION

MArtin: I started reading a bit about Dirichlet-Ferguson difusion and I still don't really understand if/how it depends on the choice of metric on the space of probability densities. It is not Brownian motion for the Otto metric (as this cant be defined), but is there some dependance on the metric? I dint un derstnad the answers that i either got by chatGPT or found in papers. Do you have a better understanding of that? E.g. would we get a different thing if we would want to construct a more Wasserstein-Fisher-Rao type Dirichlet form? Ok, actually this is done

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in Lorenzo Dello Schiavo and Giacomo Enrico Sodini (2025). The Hellinger–Kantorovich metric measure geometry on spaces of measures.

In trying to further understand these dirichlet forms constructions: Can we construct these on any infinite dimensional manifold? How would it look like in an abstract situation $\mathcal{M}, G$. Are there any particular properties of either $\mathcal{M}$ or G that we need? What about some of the standard examples, i.e.,

(1) ℓ^2 with standard ℓ^2-inner product as a metric (flat space)
(2) ℓ^2 with some conformal weighted ℓ^2-inner product (not flat anymore and no explicit geodesics)
(3) Space of curves/surfaces with reparam invariant Sob metrics
(4) Diffeomorphism group with Sobolev metrics
(5) Volume preserving diffeos with L^2-metric (i.e., Dirichlet form for the incompressible Euler equations)

Mao: My understanding is that to define a "natural" Dirichlet form (i.e., integration of the gradient), then you need a gradient on the space, and that depends on your metric. As long as you have a space with a gradient and some reference measure, you should be able to define a Dirichlet form.

2. Preliminaries

2.1. **Markov semigroups and generators.** On $\mathbb{R}^d$, Brownian motion $\{B_t\}_{t\geq 0}$ is a prototypical time-homogeneous Markov process. If $(X_t)_{t\geq 0}$ is a (time-homogeneous) Markov process on a measurable space $(E, \mathcal{B})$, its transition probabilities induce a family of linear operators $(P_t)_{t\geq 0}$ on a suitable space of functions by

$$P_t f(x) := \int_E f(y)\, p_t(x, \mathrm{d}y) := \mathbb{E}_x[f(X_t)]. \tag{2.1}$$

Under standard regularity assumptions (e.g. the Feller property on a locally compact state space [5, Theorem A.2.2]), one can also construct a Markov process (or, to be exact, a Hunt process) from such a semigroup.

If $(P_t)_{t\geq 0}$ is a strongly continuous contraction semigroup on a Hilbert space, there is a one correspondence [5, Lemma 1.3.2] with a non-positive definite self-adjoint operator L called its *generator*, defined by

$$Lf := \lim_{h\downarrow 0} \frac{P_h f - f}{h}. \tag{2.2}$$

For Brownian motion on $\mathbb{R}^d$, it is the half of the Laplacian operator $L = \frac{1}{2}\Delta$. Therefore, Brownian motion can be equivalently characterized as the Markov process with generator $\frac{1}{2}\Delta$.

A natural idea for defining a "Brownian motion" on the Wasserstein space $P_2(M)$ over a Riemannian manifold M is to use the formal Riemannian structure discovered by Otto [10] and try to build a Laplace–Beltrami operator. However, second-order calculus on $P_2(M)$ is delicate and a canonical Laplace–Beltrami operator is not available in full generality; see Gigli [6, Remark 5.6]. This motivates an alternative approach based on Dirichlet forms.

2.2. **Dirichlet forms and symmetric Markov processes.** Let $(X, \mathcal{B}, m)$ be a σ-finite measure space and consider a nonnegative definite symmetric bilinear form $\mathcal{E}$ with a dense domain $\mathcal{D}(\mathcal{E})$ on $L^2(X, m)$. It is *closed* if $\mathcal{D}(\mathcal{E})$ is complete for the norm

$$\|u\|_{\mathcal{E}_1}^2 := \|u\|_{L^2(X,m)}^2 + \mathcal{E}(u, u).$$

A *Dirichlet form* is a closed symmetric form satisfying the Markov property (stability under normal contractions), see [5, §1.4].

Closed symmetric forms correspond to non-positive self-adjoint operators: there exists a unique non-positive self-adjoint operator L on $L^2(X,m)$ such that

$$\mathcal{E}(u,v) = (-Lu, v)_{L^2(X,m)}, \qquad u \in \mathcal{D}(\mathcal{E}),\ v \in \mathcal{D}(L), \tag{2.3}$$

and conversely [5, Theorem 1.3.1]. Using the operator L, we obtain a semigroup $\{T_t\}$, and thus a Markov (or a Hunt) process. Indeed, if $(\mathcal{E},\mathcal{D}(\mathcal{E}))$ is a *regular* Dirichlet form on a locally compact separable metric space, then there exists an m-symmetric Hunt process whose Dirichlet form is $(\mathcal{E},\mathcal{D}(\mathcal{E}))$ [5, Theorem 7.2.1]. Such a processs is unique in the sense that the transition functions agree except on the *properly exceptional sets*, i.e., sets that have zero measure under m and are avoided by the process almost surely when started outside them [5, Theorem 4.2.8]

2.3. **Dirichlet–Ferguson measure.** Let M be a closed Riemannian manifold with normalized volume measure m and fix a parameter $\beta > 0$.

Definition 2.1 (Poisson–Dirichlet distribution)**.** The (one-parameter) Poisson–Dirichlet distribution $\mathrm{PD}(\beta)$ is a probability law on decreasing sequences $s = (s_i)_{i\geq 1}$ with $s_i \geq 0$ and $\sum_{i\geq 1} s_i = 1$. One way to construct it is by *stick-breaking*: take i.i.d.

$$V_1, V_2, \dots \sim \mathrm{Beta}(1,\beta),$$

set

$$\widetilde{s}_1 := V_1, \qquad \widetilde{s}_n := V_n \prod_{j=1}^{n-1}(1-V_j), \quad n \geq 2, \tag{2.4}$$

and then rank the sequence $(\widetilde{s}_n)_{n\geq 1}$ in nonincreasing order.

Remark 2.2. This construction is said to be stick-breaking because one can think of starting with a stick of length 1 and breaking off a fraction V_1 to form $\widetilde{s}_1$; then breaking off a fraction V_2 of the remaining piece to form $\widetilde{s}_2$, and so on.

Definition 2.3 (Dirichlet–Ferguson measure)**.** The *Dirichlet–Ferguson measure* (or *Dirichlet process*) with base measure m and concentration parameter β is the law on $P(M)$ of the random probability measure

$$\mu := \sum_{i=1}^{\infty} s_i\, \delta_{X_i}, \tag{2.5}$$

where $(s_i)_{i\geq 1} \sim \mathrm{PD}(\beta)$ and $(X_i)_{i\geq 1}$ are i.i.d. samples from m, independent of (s_i). We denote this law by $\mathcal{D}_m$.

By construction, $\mathcal{D}_m$-a.e. measure is purely atomic. In particular, $\mathcal{D}_m$ is supported on a dense subset of $P_2(M)$ [2, p. 596], and therefore has full support. In what follows we will mostly work on $M = \mathbb{T}^d$ equipped with its normalized volume m; in that case we simply write $\mathcal{D}$ for $\mathcal{D}_m$.

2.4. **Dirichlet–Ferguson diffusion and particle representation.** Dello Schiavo [2] constructs a $\mathcal{D}_m$-symmetric Markov process on $P(M)$ by Dirichlet form methods. We will call this process the *Dirichlet–Ferguson diffusion* A key feature (and the main reason the Dirichlet–Ferguson measure is convenient here) is the availability of a particle representation. To be more specific, the Dirichlet-Ferguson diffusion on the Wasserstein space on $\mathbb{T}^d$ can be seen as a countable superposition of independent Brownian motions on $\mathbb{T}^d$. Before introducing it formally, we need some notation. Let

$$\mathbf{T}_o^\infty := \left\{ s = (s_i)_{i\geq 1} :\ s_1 > s_2 > \dots > 0,\ \sum_{i\geq 1} s_i = 1 \right\},$$

and let $(\mathbb{T}^d)^\infty_o$ be the set of sequences $x = (x_i)_{i\geq 1}$ in $\mathbb{T}^d$ with pairwise distinct coordinates. Define the "embedding" map

$$\mathrm{em}(s,x) := \sum_{i\geq 1} s_i\,\delta_{x_i} \in P(\mathbb{T}^d). \tag{2.6}$$

Since under $\mathcal{D}$ the weights are almost surely strictly decreasing and the atoms are almost surely distinct, the image of em is a Borel set $O \subset P(\mathbb{T}^d)$ with $\mathcal{D}(O) = 1$ such that em admits a measurable right-inverse

$$\mathrm{em}^{-1} : O \to \mathbf{T}^\infty_o \times (\mathbb{T}^d)^\infty_o, \qquad \mathrm{em}(\mathrm{em}^{-1}(\mu)) = \mu.$$

(See [1, §2.3] for a concrete construction.)

2.4.1. *Canonical probability space.* Although the Dirichlet–Ferguson diffusion can be constructed abstractly via Dirichlet forms and thus do not depend on the underlying probability space, the theory we borrow from Delarue–Martini–Sodini [1] relies on a specific realization of the process. For this, we will first construct the underlying probability space which we call the *canonical probability space* for the Dirichlet–Ferguson diffusion.

Fix a horizon $T > 0$. Following [1, §2.3.1], we work on the space

$$\Xi_0 := \mathbf{T}^\infty_o \times (\mathbb{T}^d)^\infty_o \times (C([0,T];\mathbb{R}^d))^\infty,$$

with coordinates $(\varsigma, \xi_0, \beta) = ((\varsigma_i)_{i\geq 1}, (\xi_{0,i})_{i\geq 1}, (\beta_i)_{i\geq 1})$. These coordinates represent the initial data of the diffusion and the family of Brownian motions driving the particles. We then set

$$\Xi := \Xi_0 \times (C([0,T];\mathbb{R}^d))^\infty$$

and denote by $\xi = (\xi_i)_{i\geq 1}$ the last coordinate which records the particle trajectories. Let $(\mathcal{G}_t)_{t\in[0,T]}$ be the canonical filtration generated by these coordinates.

For each $t_0 \in [0,T]$ and each "mass–position" pair $(s,x) \in \mathbf{T}^\infty_o \times (\mathbb{T}^d)^\infty_o$, [1, eq. (2.29)] defines a probability measure $\mathbb{P}^{t_0,(s,x)}$ on $(\Xi, \mathcal{G}_T)$ such that under $\mathbb{P}^{t_0,(s,x)}$ the coordinates satisfy

$$\varsigma \equiv s, \qquad \xi_0 \equiv x, \qquad \xi_i(t) = x_i + \sqrt{\frac{2}{s_i}}\big(\beta_i(t) - \beta_i(t_0)\big), \quad t \in [t_0, T],\ i \geq 1, \tag{2.7}$$

and $(\beta_i)_{i\geq 1}$ are independent $\mathbb{R}^d$-Brownian motions. (As usual we interpret $\xi_i(t)$ as a $\mathbb{T}^d$-valued process via projection modulo $\mathbb{Z}^d$.) The associated measure-valued process is

$$\mu_t := \sum_{i\geq 1} \varsigma_i\,\delta_{\xi_i(t)} \in P(\mathbb{T}^d), \qquad t \in [t_0, T]. \tag{2.8}$$

When $\mu_0 \in O$ and $(s,x) = \mathrm{em}^{-1}(\mu_0)$, we write

$$\mathbb{P}^{\mathrm{em}^{-1}(\mu_0)} := \mathbb{P}^{0,(s,x)}.$$

More generally, for any $\tau \in [0,T]$ and any $\mu \in O$, if $(\varsigma, x) = \mathrm{em}^{-1}(\mu)$, we write

$$\mathbb{P}^{\tau,\mathrm{em}^{-1}(\mu)} := \mathbb{P}^{\tau,(\varsigma,x)}.$$

In particular, the notation $\mathbb{P}^{s,\mathrm{em}^{-1}(\mu)}$ used later means the law of the free particle system started from μ at time s. Under $\mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}$, the process $(\mu_t)_{t\in[0,T]}$ is the Dirichlet–Ferguson diffusion started from μ_0 [1, §2.3].

2.5. **Cylinder functions, Wasserstein gradients and Wasserstein–Sobolev spaces.** We briefly recall the functional-analytic framework of [1].

2.5.1. *Smooth cylinder functions.* A basic class of test functions on $P(\mathbb{T}^d)$ are *smooth cylinder functions* of the form

$$u(\mu) = F\big(\mu(f_1), \dots, \mu(f_k)\big), \tag{2.9}$$

where $k \in \mathbb{N}$, $F \in C_b^\infty(\mathbb{R}^k)$, $f_1, \dots, f_k \in C^\infty(\mathbb{T}^d)$ and $\mu(f) \coloneqq \int f \, \mathrm{d}\mu$. (For the Dirichlet–Ferguson measure one needs a slightly richer class allowing also "mass perturbations"; see [1, §2.2]. For the present note, (2.9) is sufficient to explain the main ideas.)

2.5.2. *Wasserstein gradient on cylinder functions.* For a cylinder function (2.9), the Otto calculus suggests that its gradient at μ is an element of the tangent space $T_\mu P_2(\mathbb{T}^d) \simeq L^2(\mathbb{T}^d, \mu; \mathbb{R}^d)$ and is given pointwise by

$$(Du)(\mu, x) \coloneqq (\nabla u)_\mu(x) = \sum_{j=1}^k \partial_j F\big(\mu(f_1), \dots, \mu(f_k)\big) \, \nabla f_j(x), \tag{2.10}$$

see [1, eq. (2.11)].

2.5.3. *Cheeger energy and $H^{1,2}$.* Let $H \coloneqq L^2(P(\mathbb{T}^d), \mathcal{D})$. The Cheeger energy $\mathrm{CE} : H \to [0, \infty]$ is defined as the $L^2(\mathcal{D})$-lower semicontinuous relaxation of the squared local Lipschitz constant with respect to W_2 [1, Definition 2.9]. The metric Sobolev space is

$$H^{1,2} \coloneqq H^{1,2}(P(\mathbb{T}^d), W_2, \mathcal{D}) \coloneqq \{u \in H : \mathrm{CE}(u) < \infty\},$$

endowed with the norm

$$\|u\|_{H^{1,2}}^2 \coloneqq \|u\|_H^2 + \mathrm{CE}(u). \tag{2.11}$$

2.5.4. *Gradient for $H^{1,2}$ functions.* Let

$$\boldsymbol{\mathcal{D}}(\mathrm{d}\mu, \mathrm{d}x) \coloneqq \big(\delta_\mu \otimes \mu\big)(\mathrm{d}\mu, \mathrm{d}x) \, \mathcal{D}(\mathrm{d}\mu)$$

be the "lift" of $\mathcal{D}$ to $P(\mathbb{T}^d) \times \mathbb{T}^d$. Delarue–Martini–Sodini prove that for each $u \in H^{1,2}$ there exists a unique

$$Du \in L^2\big(P(\mathbb{T}^d) \times \mathbb{T}^d, \boldsymbol{\mathcal{D}}; \mathbb{R}^d\big)$$

characterized by approximation by cylinder functions and such that

$$\mathrm{CE}(u, v) = \int_{P(\mathbb{T}^d) \times \mathbb{T}^d} \langle Du(\mu, x), \, Dv(\mu, x) \rangle \, \boldsymbol{\mathcal{D}}(\mathrm{d}\mu, \mathrm{d}x), \qquad u, v \in H^{1,2}, \tag{2.12}$$

see [1, Proposition 2.13].

2.5.5. *Dirichlet–Ferguson Laplacian.* The bilinear form $(u, v) \mapsto \mathrm{CE}(u, v)$ on $H^{1,2}$ is a Dirichlet form on H [1, Theorem 2.11]. Its generator is the *Dirichlet–Ferguson Laplacian* Δ with domain

$$D(\Delta) \coloneqq \big\{v \in H^{1,2} : \exists \, \Delta v \in H \text{ such that } \mathrm{CE}(u, v) = -(u, \Delta v)_H \ \forall u \in H^{1,2}\big\},$$

compare [1, eq. (2.24)].

2.5.6. *Chain rule.* We will repeatedly use the usual chain rule for Sobolev functions.

Proposition 2.4 (Chain rule, Proposition 5.2(e), [3])**.** *If $u \in H^{1,2}$ and $\phi : \mathbb{R} \to \mathbb{R}$ is C^1 with bounded derivative, then $\phi \circ u \in H^{1,2}$ and*

$$D(\phi \circ u) = \phi'(u) \, Du, \qquad \boldsymbol{\mathcal{D}}\text{-a.e.} \tag{2.13}$$

In particular, if $u > 0$ and log is applied on a range where it is Lipschitz, then $D \log u = Du/u$.

3. Infinite dimensional conditioning via Doob h-transform

We keep the notation of [1]. Set

$$H := L^2\big(P(\mathbb{T}^d), \mathcal{D}\big), \qquad H^{1,2} := H^{1,2}\big(P(\mathbb{T}^d), W_2, \mathcal{D}\big). \tag{3.1}$$

Theorem 3.1 (Heat equation and martingale representation)**.** *Let $g \in H$ and $T > 0$. There exists a unique*

$$u \in C([0,T]; H) \cap \mathrm{AC}_{loc}((0,T]; H) \cap L^2([0,T]; H^{1,2})$$

such that $u_t \in D(\Delta)$ for a.e. $t \in (0,T)$ and

$$\begin{cases} \partial_t u_t + \Delta u_t = 0, & \textit{for a.e. } t \in (0,T), \\ u_T = g. \end{cases} \tag{3.2}$$

Moreover, for any initial condition $\mu_0 \in O$ and the Dirichlet–Ferguson diffusion $(\mu_t)_{t\in[0,T]}$ constructed on $(\Xi, \mathcal{G}_T, \mathbb{P}^{\mathrm{em}^{-1}(\mu_0)})$ by (2.8), we have the martingale representation

$$u_t(\mu_t) = u_0(\mu_0) + \sum_{i=1}^{\infty} \int_0^t \sqrt{2\varsigma_i}\, Du_r(\mu_r, \xi_i(r)) \cdot \mathrm{d}\beta_i(r), \qquad t \in [0,T], \quad \mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}\text{-a.s.} \tag{3.3}$$

where $(\beta_i)_{i\geq 1}$ are the coordinate Brownian motions on Ξ.

Proof. The well-posedness of (3.2) is Corollary 3.5 in [1] (for the case $b = 0$ and $f = 0$). For the martingale representation, Proposition 4.21 in [1] yields, for $0 \leq t \leq T$,

$$g(\mu_T) = u_t(\mu_t) + \sum_{i=1}^{\infty} \int_t^T \sqrt{2\varsigma_i}\, Du_r(\mu_r, \xi_i(r)) \cdot \mathrm{d}\beta_i(r), \qquad \mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}\text{-a.s.}$$

Evaluating at $t = 0$ and a general t and subtracting the two identities gives (3.3). □

3.1. A Doob h-transform via Girsanov.

To define $\log u_t(\mu_t)$ we assume strict positivity.

Assumption 3.2. The terminal datum $g \in H \cap L^\infty(\mathcal{D})$ is bounded and satisfies $g \geq c > 0$.

We fix, for each $t \in [0,T]$, the Borel representative of u_t furnished by Theorem 4.19 in [1]. For this representative, the martingale formula in Theorem 3.1 yields

$$u_t(\mu_t) = \mathbb{E}^{\mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}}\big[g(\mu_T) \mid \mathcal{G}_t\big] \geq c, \qquad t \in [0,T], \quad \mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}\text{-a.s.}$$

In particular, the pathwise quantity $u_t(\mu_t)$ is well defined for the chosen Borel representative and is strictly positive for all $t \in [0,T]$.

Let

$$h_t := u_t(\mu_t), \qquad Y_t := \log h_t.$$

By (3.3), $(h_t)_{t\in[0,T]}$ is a continuous square-integrable martingale under $\mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}$. More precisely,

$$\mathrm{d}h_t = \sum_{i=1}^{\infty} \sqrt{2\varsigma_i}\, Du_t(\mu_t, \xi_i(t)) \cdot \mathrm{d}\beta_i(t), \tag{3.4}$$

and its quadratic variation is

$$\mathrm{d}[h]_t = \sum_{i=1}^{\infty} 2\varsigma_i \, \|Du_t(\mu_t, \xi_i(t))\|^2 \,\mathrm{d}t. \tag{3.5}$$

Indeed, the cross-variations vanish because $(\beta_i)_{i\geq 1}$ have independent components, and the series is summable thanks to [1, eq. (4.54)]. Check

Applying Itô's formula to the real semimartingale h_t and the function log yields

$$\mathrm{d}Y_t = \frac{1}{h_t}\,\mathrm{d}h_t - \frac{1}{2h_t^2}\,\mathrm{d}[h]_t. \tag{3.6}$$

Using (3.5) and the chain rule (2.13) (so that $D\log u_t = Du_t/u_t$), we obtain

$$\mathrm{d}Y_t = \sum_{i=1}^{\infty} \sqrt{2\varsigma_i}\, D\log u_t(\mu_t, \xi_i(t))\cdot \mathrm{d}\beta_i(t) - \sum_{i=1}^{\infty} \varsigma_i\, \|D\log u_t(\mu_t,\xi_i(t))\|^2\,\mathrm{d}t. \tag{3.7}$$

Set

$$Z_t := \exp(Y_t - Y_0) = \frac{u_t(\mu_t)}{u_0(\mu_0)}.$$

Since (h_t) is a positive martingale and μ_0 is deterministic, $\mathbb{E}^{\mathbb{P}^{\mathrm{em}^{-1}}(\mu_0)}[Z_t] = 1$. Writing $\beta_t := (\beta_i(t))_{i\geq 1}$ and viewing it as a Brownian motion in the Hilbert space $\ell^2(\mathbb{N};\mathbb{R}^d)$, (3.7) shows that Z is the Doléans–Dade exponential of the stochastic integral with integrand

$$\sigma_t := \left(\sqrt{2\varsigma_1}\, D\log u_t(\mu_t,\xi_1(t)), \sqrt{2\varsigma_2}\, D\log u_t(\mu_t,\xi_2(t)), \dots\right) \in \ell^2(\mathbb{N};\mathbb{R}^d). \tag{3.8}$$

Define a new probability measure $\widetilde{\mathbb{P}}$ on $(\Xi, \mathcal{G}_T)$ by

$$\frac{\mathrm{d}\widetilde{\mathbb{P}}}{\mathrm{d}\mathbb{P}^{\mathrm{em}^{-1}(\mu_0)}}\Big|_{\mathcal{G}_T} = Z_T = \frac{g(\mu_T)}{u_0(\mu_0)}. \tag{3.9}$$

By Girsanov's theorem in Hilbert spaces [8, Theorem I.0.6], the processes

$$\widehat{\beta}_i(t) := \beta_i(t) - \int_0^t \sqrt{2\varsigma_i}\, D\log u_r(\mu_r, \xi_i(r))\,\mathrm{d}r \tag{3.10}$$

are independent Brownian motions under $\widetilde{\mathbb{P}}$. Consequently, under $\widetilde{\mathbb{P}}$ each particle satisfies the SDE

$$\mathrm{d}\xi_i(t) = \sqrt{\frac{2}{\varsigma_i}}\,\mathrm{d}\widehat{\beta}_i(t) + 2\, D\log u_t(\mu_t,\xi_i(t))\,\mathrm{d}t, \tag{3.11}$$

and the induced measure-valued process $\mu_t = \sum_{i\geq 1}\varsigma_i\delta_{\xi_i(t)}$ solves a drifted version of the Dirichlet–Ferguson diffusion. The change of measure (3.9) can be interpreted as a Doob h-transform (or conditioning by terminal weight) since it reweights paths according to $g(\mu_T)$.

3.2. **Numerical simulation.** We now explain how the finite-dimensional h-transformed dynamics (4.11) can be simulated numerically once the mass vector $s \in \Sigma_n^{1/n}$ has been frozen. For a fixed s, write

$$b_i^{(n)}(t,x;s) := 2D_i^{(n)}\log u_t^{(n)}(s,x) = \frac{2}{s_i}\nabla_{x_i}\log u_t^{(n)}(s,x), \qquad i = 1,\dots,n. \tag{3.12}$$

Then (4.11) becomes the time-inhomogeneous SDE

$$\mathrm{d}\xi_i^{(n)}(t) = b_i^{(n)}(t,\xi^{(n)}(t);s)\,\mathrm{d}t + \sqrt{\frac{2}{s_i}}\,\mathrm{d}\widehat{\beta}_i^{(n)}(t), \qquad i=1,\dots,n. \tag{3.13}$$

Since $u_t^{(n)} = P_{T-t}^{(n)} g_n$ and the semigroup acts only in the torus variables, one has the heat-kernel representation

$$u_t^{(n)}(s,x) = \int_{(\mathbb{T}^d)^n} g_n(s,y)\prod_{i=1}^n p_{(T-t)/s_i}^{\mathbb{T}^d}(x_i,y_i)\,\mathrm{d}y, \tag{3.14}$$

where $p_\tau^{\mathbb{T}^d}$ denotes the heat kernel on $\mathbb{T}^d$ corresponding to the equation $\partial_\tau u = \Delta u$. Thus the numerical task is to evaluate $u_t^{(n)}$ and its logarithmic gradient and then to discretize (3.13).

Fix a time grid $0 = t_0 < t_1 < \dots < t_M = T$ with $\Delta t = T/M$. Writing $X^m = (X_1^m,\dots,X_n^m)$ for the numerical approximation to $\xi^{(n)}(t_m)$, the Euler–Maruyama update is

$$X_i^{m+1} = X_i^m + b_i^{(n)}(t_m, X^m; s)\,\Delta t + \sqrt{\frac{2\Delta t}{s_i}}\, Z_i^m, \qquad Z_i^m \sim N(0, I_d), \tag{3.15}$$

followed by the projection of X_i^{m+1} back to $\mathbb{T}^d$ modulo $\mathbb{Z}^d$. The associated measure-valued approximation is then

$$\mu_{t_m}^{(n)} = \sum_{i=1}^{n} s_i \, \delta_{X_i^m}.$$

For a generic terminal condition g_n, the main computational cost is the evaluation of (3.14). This can be done by methods such as Fourier methods on the torus, or Monte Carlo evaluation of the free particle semigroup. The scheme becomes especially simple when g_n is chosen so that $u_t^{(n)}$ is explicit.

Gaussian terminal condition centred at a prescribed configuration. Because the finite-dimensional dynamics only move the particle locations, the natural way to promote the terminal measure toward a prescribed atomic target is to keep the mass vector s fixed and choose a terminal weight concentrated around a target configuration $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n) \in (\mathbb{T}^d)^n$. The corresponding target measure is

$$\bar{\mu}^{(n)} \coloneqq \sum_{i=1}^{n} s_i \, \delta_{\bar{x}_i}.$$

A convenient weighted Gaussian terminal condition on the universal cover $(\mathbb{R}^d)^n$ is

$$g_{n,\lambda}(s, x) \coloneqq \exp\left(-\frac{\lambda}{2} \sum_{i=1}^{n} s_i \, |x_i - \bar{x}_i|^2 \right), \qquad \lambda > 0. \tag{3.16}$$

This is the natural Gaussian centred at the configuration $(s, \bar{x})$ for the weighted geometry induced by the generator $L_s^{(n)} = \sum_i s_i^{-1} \Delta_{x_i}$. On the torus one may either periodize (3.16), or work on a fixed lift as long as wrap-around effects are negligible on the time interval of interest.

We now derive the corresponding closed form for $u_t^{(n)}$ in detail. Set

$$\tau \coloneqq T - t, \qquad v_\tau^{(n)}(s, x) \coloneqq u_{T-\tau}^{(n)}(s, x).$$

Since $u_t^{(n)}$ solves the backward equation

$$\partial_t u_t^{(n)} + \sum_{i=1}^{n} \frac{1}{s_i} \Delta_{x_i} u_t^{(n)} = 0, \qquad u_T^{(n)} = g_{n,\lambda},$$

the function $v_\tau^{(n)}$ solves the forward heat equation

$$\partial_\tau v_\tau^{(n)} = \sum_{i=1}^{n} \frac{1}{s_i} \Delta_{x_i} v_\tau^{(n)}, \qquad v_0^{(n)} = g_{n,\lambda}. \tag{3.17}$$

Because both the generator and the terminal datum factorize over the particle labels, one expects the solution to keep the same Gaussian shape. Writing

$$R(x) \coloneqq \sum_{i=1}^{n} s_i \, |x_i - \bar{x}_i|^2,$$

we therefore make the ansatz

$$v_\tau^{(n)}(s, x) = A(\tau) \exp\left(-\frac{B(\tau)}{2} R(x) \right), \tag{3.18}$$

with initial conditions $A(0) = 1$ and $B(0) = \lambda$ so that $v_0^{(n)} = g_{n,\lambda}$.

Differentiating (3.18) with respect to τ gives

$$\partial_\tau v_\tau^{(n)} = \left(\frac{A'(\tau)}{A(\tau)} - \frac{B'(\tau)}{2} R(x) \right) v_\tau^{(n)}.$$

For each $i = 1, \dots, n$ we also have

$$\nabla_{x_i} v_\tau^{(n)} = -B(\tau) s_i (x_i - \bar{x}_i) v_\tau^{(n)},$$

and therefore

$$\Delta_{x_i} v_\tau^{(n)} = \Big(-B(\tau) s_i d + B(\tau)^2 s_i^2 |x_i - \bar{x}_i|^2\Big) v_\tau^{(n)}.$$

Summing with the weights $1/s_i$ from the generator gives

$$\sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_\tau^{(n)} = \left(-nd\, B(\tau) + B(\tau)^2 \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2\right) v_\tau^{(n)} = \big(-nd\, B(\tau) + B(\tau)^2 R(x)\big) v_\tau^{(n)}.$$

Substituting this identity and the expression for $\partial_\tau v_\tau^{(n)}$ into (3.17) and identifying the coefficients of 1 and $R(x)$ yields the ODE system

$$\frac{A'(\tau)}{A(\tau)} = -nd\, B(\tau), \qquad -\frac{1}{2} B'(\tau) = B(\tau)^2, \qquad A(0) = 1, \quad B(0) = \lambda. \tag{3.19}$$

The second ODE is scalar and can be solved explicitly:

$$B'(\tau) = -2B(\tau)^2 \qquad \Longrightarrow \qquad B(\tau) = \frac{\lambda}{1 + 2\lambda\tau}.$$

Plugging this into the first ODE gives

$$\log A(\tau) = -nd \int_0^\tau \frac{\lambda}{1 + 2\lambda r}\, \mathrm{d}r = -\frac{nd}{2} \log(1 + 2\lambda\tau),$$

so that

$$A(\tau) = (1 + 2\lambda\tau)^{-nd/2}.$$

Substituting A and B back into (3.18) and recalling that $\tau = T - t$ yields the closed formula

$$u_t^{(n)}(s, x) = \big(1 + 2\lambda(T - t)\big)^{-nd/2} \exp\left(-\frac{\lambda}{2(1 + 2\lambda(T - t))} \sum_{i=1}^n s_i\, |x_i - \bar{x}_i|^2\right). \tag{3.20}$$

As a consistency check, when $t = T$ the prefactor equals 1 and (3.20) reduces exactly to the terminal condition (3.16).

Taking logarithms in (3.20),

$$\log u_t^{(n)}(s, x) = -\frac{nd}{2} \log\big(1 + 2\lambda(T - t)\big) - \frac{\lambda}{2(1 + 2\lambda(T - t))} \sum_{i=1}^n s_i |x_i - \bar{x}_i|^2.$$

Hence, for each i,

$$\nabla_{x_i} \log u_t^{(n)}(s, x) = -\frac{\lambda s_i}{1 + 2\lambda(T - t)} (x_i - \bar{x}_i),$$

and therefore

$$D_i^{(n)} \log u_t^{(n)}(s, x) = \frac{1}{s_i} \nabla_{x_i} \log u_t^{(n)}(s, x) = -\frac{\lambda}{1 + 2\lambda(T - t)} (x_i - \bar{x}_i), \qquad b_i^{(n)}(t, x; s) = -\frac{2\lambda}{1 + 2\lambda(T - t)} (x_i - \bar{x}_i). \tag{3.21}$$

Thus each particle satisfies the time-inhomogeneous Ornstein–Uhlenbeck-type SDE

$$\mathrm{d}\xi_i^{(n)}(t) = -\frac{2\lambda}{1 + 2\lambda(T - t)} \big(\xi_i^{(n)}(t) - \bar{x}_i\big)\, \mathrm{d}t + \sqrt{\frac{2}{s_i}}\, \mathrm{d}\widehat{\beta}_i^{(n)}(t). \tag{3.22}$$

For this weighted Gaussian choice, the restoring drift is independent of s_i, while the noise intensity still depends on $s_i^{-1/2}$.

Algorithm 1 Euler–Maruyama scheme for the Gaussian terminal condition centred at $\bar{\mu}^{(n)}$

Require: n, masses $s = (s_1, \dots, s_n)$, target locations $\bar{x} = (\bar{x}_1, \dots, \bar{x}_n)$, parameter $\lambda > 0$, horizon T, step size Δt, and initial configuration $X^0 \in (\mathbb{T}^d)^n$

Ensure: Approximate path $(X^m)_{m=0}^M$ and measures $\mu_{t_m}^{(n)} = \sum_{i=1}^n s_i \delta_{X_i^m}$

$M \leftarrow T/\Delta t$ and $t_m \leftarrow m\Delta t$ for $m = 0, \dots, M$

for $m = 0, \dots, M-1$ **do**

  $\tau_m \leftarrow T - t_m$

  for $i = 1, \dots, n$ **do**

    Either use the Euclidean closed form

$$b_i^m \leftarrow -\frac{2\lambda}{1 + 2\lambda\tau_m}(X_i^m - \bar{x}_i),$$

    or use the wrapped drift obtained from a truncated torus image sum

    Draw $Z_i^m \sim N(0, I_d)$ independently

    $X_i^{m+1} \leftarrow X_i^m + b_i^m \, \Delta t + \sqrt{\frac{2\Delta t}{s_i}} \, Z_i^m$

    Reduce X_i^{m+1} modulo $\mathbb{Z}^d$

  end for

end for

4. Lagrangian Approximation

We will now consider the finite dimensional approximation of the diffusion with drift constructed in the previous subsection. Our main goal is to complete the following program:

(1) Identify the transition family associated with the drifted process.
(2) Describe the finite-dimensional approximation of the drifted process and identify its transition family.
(3) Prove that the finite-dimensional transition families converge to the infinite-dimensional one.

4.0.1. *The transition family of the drifted process.* More precisely, since the drifted dynamics are time-inhomogeneous, the natural object is a two-parameter transition family. For $0 \le s \le T$ and $\mu \in O$, let $\widetilde{\mathbb{P}}^{s,\mu}$ be the probability measure on $(\Xi, \mathcal{G}_T)$ defined by

$$\frac{\mathrm{d}\widetilde{\mathbb{P}}^{s,\mu}}{\mathrm{d}\mathbb{P}^{s,\mathrm{em}^{-1}(\mu)}}\Big|_{\mathcal{G}_T} = \frac{g(\mu_T)}{u_s(\mu)}. \tag{4.1}$$

When $s = 0$ and $\mu = \mu_0$, this coincides with the measure $\widetilde{\mathbb{P}}$ introduced above. Under $\widetilde{\mathbb{P}}^{s,\mu}$, the coordinate process solves the drifted particle system

$$\mu_t := \sum_{i \ge 1} \varsigma_i \, \delta_{\xi_i(t)}, \qquad \mathrm{d}\xi_i(t) = \sqrt{\frac{2}{\varsigma_i}} \, \mathrm{d}\widehat{\beta}_i(t) + 2\, D \log u_t(\mu_t, \xi_i(t)) \, \mathrm{d}t, \qquad t \in [s, T]. \tag{4.2}$$

We define its transition family by

$$(\tilde{P}_{s,t} f)(\mu) := \mathbb{E}^{\widetilde{\mathbb{P}}^{s,\mu}}\big[f(\mu_t)\big], \qquad 0 \le s \le t \le T.$$

Proposition 4.1. *For every $0 \le s \le t \le T$ and $f \in H$,*

$$(\tilde{P}_{s,t} f)(\mu) = \frac{1}{u_s(\mu)} P_{t-s}(u_t f)(\mu). \tag{4.3}$$

Proof. By the definition of $\widetilde{\mathbb{P}}^{s,\mu}$,

$$\mathbb{E}^{\widetilde{\mathbb{P}}^{s,\mu}}\big[f(\mu_t)\big] = \frac{1}{u_s(\mu)} \mathbb{E}^{\mathbb{P}^{s,\mathrm{em}^{-1}(\mu)}}\big[f(\mu_t) g(\mu_T)\big].$$

Using the Markov property at time t and the Borel representative of u_t fixed above, we obtain

$$\mathbb{E}^{\mathbb{P}^{s,\mathrm{em}^{-1}(\mu)}}\big[g(\mu_T) \mid \mathcal{G}_t\big] = u_t(\mu_t),$$

hence

$$\mathbb{E}^{\mathbb{P}^{s,\mathrm{em}^{-1}(\mu)}}\big[f(\mu_t)g(\mu_T)\big] = \mathbb{E}^{\mathbb{P}^{s,\mathrm{em}^{-1}(\mu)}}\big[f(\mu_t)u_t(\mu_t)\big] = P_{t-s}(u_t f)(\mu).$$

This proves the claim. □

Remark 4.2. The expression $P_{t-s}(u_t f)$ is well defined for each $f \in H$ because $u_t = P_{T-t}g$ is bounded by $\|g\|_{L^\infty}$.

4.0.2. *Finite-dimensional approximation of the free process.* We now recall the finite-dimensional approximation used in [2, Section 6.1]. Let

$$T := \Big\{ s = (s_i)_{i\geq 1} \in [0,1]^{\mathbb{N}} : \sum_{i=1}^{\infty} s_i = 1,\ s_1 \geq s_2 \geq \cdots \geq 0 \Big\},$$

and let

$$T^\circ := \Big\{ s \in T : s_1 > s_2 > \cdots > 0 \Big\}.$$

For each $n \in \mathbb{N}$, define the coordinate projection

$$\mathrm{pr}_n : T \to \mathbb{R}^n, \qquad \mathrm{pr}_n(s_1, s_2, \dots) = (s_1, \dots, s_n).$$

We now define

$$\Sigma_n := \mathrm{pr}_n(T), \qquad \Sigma_n^{1/n} := \{ s = (s_1, \dots, s_n) \in \Sigma_n : s_1 \geq 1/n \},$$

and define

$$\widehat{M}_n := \Sigma_n^{1/n} \times (\mathbb{T}^d)^n.$$

We endow $\Sigma_n^{1/n}$ with the measure

$$\pi_n^\beta := (\mathrm{pr}_n)_{\#}\mathrm{PD}(\beta)\big|_{\Sigma_n^{1/n}},$$

and $\widehat{M}_n$ with the product measure

$$\widehat{m}_n := \pi_n^\beta \otimes m^{\otimes n}.$$

We also define the map

$$\Phi_n : \widehat{M}_n \to \mathcal{P}(\mathbb{T}^d), \qquad \Phi_n(s,x) := \sum_{i=1}^{n} s_i \delta_{x_i}.$$

For each fixed $s \in \Sigma_n^{1/n}$, let

$$W_t^{n,s} = (\xi^1(t), \dots, \xi^n(t)), d\xi^i(t) = \sqrt{\frac{2}{s_i}}\, d\beta_i(t), \qquad t \in [0,T],\ i = 1, \dots, n,$$

where $\beta_1, \dots, \beta_n$ are independent Brownian motions on $\mathbb{T}^d$. Equivalently, the generator on the fiber $(\mathbb{T}^d)^n$ is

$$L^{(n),s} = \sum_{i=1}^{n} s_i^{-1} \Delta_{x_i}.$$

The corresponding measure-valued process is

$$\mu_t^{(n)} = \Phi_n(s, W_t^{n,s}) = \sum_{i=1}^{n} s_i \delta_{\xi_t^i}.$$

We will consider the Hilbert space for the finite dimensional approximation to be

$$H_n := L^2(\widehat{M}_n, \widehat{m}_n).$$

The free n-particle system naturally lives on $\widehat{M}_n$: the mass vector is frozen and only the position variable evolves. Accordingly, if $f \in H_n$ and $(r,x) \in \widehat{M}_n$, we define

$$(P_t^{(n)} f)(r,x) := \mathbb{E}\big[f(r, W_t^{n,r}) \mid W_0^{n,r} = x\big]. \tag{4.4}$$

Its generator acts only on the torus variables and is given by

$$L^{(n)} = \sum_{i=1}^n r_i^{-1} \Delta_{x_i}.$$

CHECK For a function $g_n \in H_n$ with $g_n \geq c > 0$, let $u_t^{(n)}$ be the solution to the finite-dimensional backward heat equation

$$\begin{cases} \partial_t u_t^{(n)} + L^{(n)} u_t^{(n)} = 0, \quad t \in (0,T), \\ u_T^{(n)} = g_n. \end{cases} \tag{4.5}$$

To do calculus on $\widehat{M}_n$, we keep the mass variable fixed and differentiate only in the torus directions. In exact analogy with the infinite-dimensional construction, the gradient must therefore be defined first on smooth functions and then extended by closure to the corresponding Sobolev space.

We introduce the weighted gradient space

$$\mathfrak{H}_n := L^2\big(\widehat{M}_n, \widehat{m}_n; (\mathbb{R}^d)^n\big), \qquad \|V\|_{\mathfrak{H}_n}^2 := \int_{\widehat{M}_n} \sum_{i=1}^n r_i |V_i(r,x)|^2 \, \widehat{m}_n(\mathrm{d}r, \mathrm{d}x).$$

For $u \in C^1(\widehat{M}_n)$ we define the smooth fiber-gradient

$$D_{\mathrm{sm}}^{(n)} u(r,x) := \Big(\frac{1}{r_1} \nabla_{x_1} u(r,x), \dots, \frac{1}{r_n} \nabla_{x_n} u(r,x)\Big), \tag{4.6}$$

where the values on the $\widehat{m}_n$-negligible set $\{r_i = 0\}$ are irrelevant. The associated quadratic form is

$$\mathcal{E}_n^{\mathrm{sm}}(u,v) := \int_{\widehat{M}_n} \sum_{i=1}^n r_i \left\langle D_{\mathrm{sm},i}^{(n)} u(r,x),\, D_{\mathrm{sm},i}^{(n)} v(r,x) \right\rangle \widehat{m}_n(\mathrm{d}r,\mathrm{d}x) = \int_{\widehat{M}_n} \sum_{i=1}^n \frac{1}{r_i} \left\langle \nabla_{x_i} u(r,x),\, \nabla_{x_i} v(r,x) \right\rangle \widehat{m}_n(\mathrm{d}r,\mathrm{d}x). \tag{4.7}$$

This is the finite-dimensional counterpart of the identity (2.12).

Proposition 4.3 (Finite-dimensional Sobolev space and gradient)**.** *The form $(\mathcal{E}_n^{\mathrm{sm}}, C^1(\widehat{M}_n))$ is closable on H_n. We denote its closure by $(\mathcal{E}_n, H_n^{1,2})$ and endow $H_n^{1,2}$ with the norm*

$$\|u\|_{H_n^{1,2}}^2 := \|u\|_{H_n}^2 + \mathcal{E}_n(u,u).$$

The operator $D_{\mathrm{sm}}^{(n)} : C^1(\widehat{M}_n) \to \mathfrak{H}_n$ is closable. Its closure, still denoted by

$$D^{(n)} : H_n^{1,2} \longrightarrow \mathfrak{H}_n,$$

is characterized as follows: for every $u \in H_n^{1,2}$ there exists a sequence $u_k \in C^1(\widehat{M}_n)$ such that

$$u_k \to u \text{ in } H_n, \qquad D_{\mathrm{sm}}^{(n)} u_k \to D^{(n)} u \text{ in } \mathfrak{H}_n,$$

and the limit $D^{(n)} u$ does not depend on the approximating sequence. Moreover,

$$\mathcal{E}_n(u,v) = \left\langle D^{(n)} u,\, D^{(n)} v \right\rangle_{\mathfrak{H}_n}, \qquad u,v \in H_n^{1,2}. \tag{4.8}$$

Proof. The closability of $(\mathcal{E}_n^{\mathrm{sm}}, C^1(\widehat{M}_n))$ is precisely Proposition 6.2 in [2]. On $C^1(\widehat{M}_n)$ we have $\mathcal{E}_n^{\mathrm{sm}}(u) = \|D_{\mathrm{sm}}^{(n)} u\|_{\mathfrak{H}_n}^2$. Hence, if $u_k \in C^1(\widehat{M}_n)$ satisfies $u_k \to 0$ in H_n and $D_{\mathrm{sm}}^{(n)} u_k \to V$ in $\mathfrak{H}_n$, then (u_k) is $\mathcal{E}_n^{\mathrm{sm}}$-Cauchy; by closability, $\mathcal{E}_n^{\mathrm{sm}}(u_k,u_k) \to 0$, so $\|V\|_{\mathfrak{H}_n} = 0$. Therefore $D_{\mathrm{sm}}^{(n)}$ is closable. The uniqueness of the limit follows from the standard argument of the closability. Finally, the identity (4.8) follows from the polarization from the equality $\mathcal{E}_n(u,u) = \|D^{(n)}u\|^2_{\mathfrak{H}_n}$.

□

Now fix $(s,x) \in \widehat{M}_n$ and define

$$h_t^{(n)} \coloneqq u_t^{(n)}(s, W_t^{n,s}).$$

This process satisfies the following properties.

Proposition 4.4. *(1) $h_t^{(n)} \geq c$.*

(2) $h_t^{(n)}$ is a martingale and satisfies

$$\mathrm{d}h_t^{(n)} = \sum_{i=1}^n \sqrt{2s_i}\, D_i^{(n)} u_t^{(n)}(s, W_t^{n,s}) \cdot \mathrm{d}\beta_i(t). \tag{4.9}$$

Proof. For the lower bound, use the positivity preserving property of the semigroup $P_t^{(n)}$:

$$u_t^{(n)} = P_{T-t}^{(n)} g_n \geq c\, P_{T-t}^{(n)} \mathbf{1} = c,$$

hence $h_t^{(n)} = u_t^{(n)}(s, W_t^{n,s}) \geq c$ for every $t \in [0,T]$.

For the martingale statement, fix the mass vector s and write

$$v_n(t,z) \coloneqq u_t^{(n)}(s,z), \qquad z = (z_1, \ldots, z_n) \in (\mathbb{T}^d)^n.$$

Then v_n solves

$$\partial_t v_n + \sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_n = 0.$$

Applying Itô's formula to $t \mapsto v_n(t, W_t^{n,s}) = h_t^{(n)}$ gives Check ito and smoothness. gradient doesnt work for non smooth vn

$$\begin{aligned}
\mathrm{d}h_t^{(n)} &= \left(\partial_t v_n + \sum_{i=1}^n \frac{1}{s_i} \Delta_{x_i} v_n \right)(t, W_t^{n,s})\, \mathrm{d}t + \sum_{i=1}^n \sqrt{\frac{2}{s_i}}\, \nabla_{x_i} v_n(t, W_t^{n,s}) \cdot \mathrm{d}\beta_i(t) \\
&= \sum_{i=1}^n \sqrt{\frac{2}{s_i}}\, \nabla_{x_i} u_t^{(n)}(s, W_t^{n,s}) \cdot \mathrm{d}\beta_i(t) \\
&= \sum_{i=1}^n \sqrt{2s_i}\, D_i^{(n)} u_t^{(n)}(s, W_t^{n,s}) \cdot \mathrm{d}\beta_i(t).
\end{aligned}$$

Hence $h_t^{(n)}$ is a local martingale. The uniform lower bound and boundedness of $u_t^{(n)}$ imply square-integrability on every finite time interval, so the local martingale is in fact a martingale.

If one starts from merely bounded terminal data g_n, the same identity follows by approximating g_n by smooth terminal conditions and passing to the limit; we omit this standard regularization argument.

□

By following the same argument as in the infinite-dimensional case, the density process for the finite-dimensional h-transform is built from

$$Z_t^{(n)} \coloneqq \frac{u_t^{(n)}(s, W_t^{n,s})}{u_0^{(n)}(s,x)},$$

and the corresponding Girsanov shift is

$$\widehat{\beta}_i^{(n)}(t) = \beta_i(t) - \int_0^t \sqrt{2s_i}\, D_i^{(n)} \log u_r^{(n)}(s, W_r^{n,s})\, \mathrm{d}r. \tag{4.10}$$

Consequently,

$$\mathrm{d}\xi_i^{(n)}(t) = \sqrt{\frac{2}{s_i}}\mathrm{d}\widehat{\beta}_i^{(n)}(t) + 2D_i^{(n)} \log u_t^{(n)}(s, W_t^{n,s})\,\mathrm{d}t. \tag{4.11}$$

4.0.3. *The transition family of the finite-dimensional approximation and its convergence.* Similar to the infinite-dimensional case, for $0 \le \tau \le t \le T$ and $(r,x) \in \widehat{M}_n$ let $\mathbb{P}^{(n),\tau,(r,x)}$ denote the law of the free finite-dimensional particle system started from (r,x) at time τ, and let $\widetilde{\mathbb{P}}^{(n),\tau,(r,x)}$ be its h-transform defined by

$$\frac{\mathrm{d}\widetilde{\mathbb{P}}^{(n),\tau,(r,x)}}{\mathrm{d}\mathbb{P}^{(n),\tau,(r,x)}} = \frac{g_n(r, W_T^{n,r})}{u_\tau^{(n)}(r,x)}.$$

We then set

$$(\tilde{P}_{\tau,t}^{(n)} f)(r,x) := \mathbb{E}^{\widetilde{\mathbb{P}}^{(n),\tau,(r,x)}}\big[f(r, W_t^{n,r})\big].$$

Proposition 4.5. *For every* $0 \le \tau \le t \le T$*,*

$$(\tilde{P}_{\tau,t}^{(n)} f)(r,x) = \frac{1}{u_\tau^{(n)}(r,x)} P_{t-\tau}^{(n)}(u_t^{(n)} f)(r,x). \tag{4.12}$$

Proof. The proof is the same Bayes/Markov argument as in the infinite-dimensional case. □

Finally, we will show that $\tilde{P}_{s,t}^{(n)}$ converges to $\tilde{P}_{s,t}$ as $n \to \infty$. We will first define the mode of the convergence.

Definition 4.6 (Generalized Hilbert convergence)**.** Following [2, Appendix A.3], let H and $(H_n)_{n\in\mathbb{N}}$ be Hilbert spaces and let $\mathcal{H} = H \coprod \bigsqcup_{n\in\mathbb{N}} H_n$. Let $D \subset H$ be dense, and let $\Lambda_n : D \to H_n$ be densely defined linear maps. We say that H_n *$\mathcal{H}$-converges* to H if

$$\|\Lambda_n u\|_{H_n} \xrightarrow[n\to\infty]{} \|u\|_H \qquad \text{for all } u \in D.$$

A sequence $u_n \in H_n$ is said to *$\mathcal{H}$-strongly converge* to $u \in H$ if there exists a sequence $\tilde{u}^m \in D$ such that

$$\|\tilde{u}^m - u\|_H \xrightarrow[m\to\infty]{} 0, \qquad \lim_{m\to\infty} \limsup_{n\to\infty} \|\Lambda_n \tilde{u}^m - u_n\|_{H_n} = 0.$$

A sequence $u_n \in H_n$ is said to *$\mathcal{H}$-weakly converge* to $u \in H$ if for every sequence $v_n \in H_n$ that $\mathcal{H}$-strongly converges to some $v \in H$ one has

$$\langle u_n, v_n\rangle_{H_n} \xrightarrow[n\to\infty]{} \langle u, v\rangle_H.$$

Let $B_n \in \mathcal{B}(H_n)$ be bounded operators and let $B \in \mathcal{B}(H)$. We say that B_n *converges $\mathcal{H}$-strongly* to B if for every sequence $u_n \in H_n$ that $\mathcal{H}$-strongly converges to $u \in H$, the sequence $B_n u_n$ $\mathcal{H}$-strongly converges to Bu.

We now map this definition into our setting. Our limiting Hilbert space is $H = L^2(P(\mathbb{T}^d), \mathcal{D})$, while the approximating Hilbert spaces are the previously defined spaces $H_n = L^2(\widehat{M}_n, \widehat{m}_n)$. We will prove a convenient representation of H_n as a closed subspace of H and identify the corresponding orthogonal projection. For this, we introduce the following class of functions on $P(\mathbb{T}^d)$: for $\hat{f} \in C_c^\infty(\mathbb{T}^d \times (1/n, 1])$, define

$$\hat{f}^\star(\eta) := \int_{\mathbb{T}^d} \hat{f}\big(x, \eta(\{x\})\big)\, \eta(\mathrm{d}x),$$

and let $\mathcal{B}_{1/n}$ be the σ-algebra generated by the cylinder functions

$$F\big(\hat{f}_1^\star, \dots, \hat{f}_k^\star\big), \qquad F \in C_b^\infty(\mathbb{R}^k),\ \hat{f}_j \in C_c^\infty(\mathbb{T}^d \times (1/n, 1]).$$

We also set

$$H_{1/n} := L^2\big(P(\mathbb{T}^d), \mathcal{B}_{1/n}, \mathcal{D}\big), \qquad P_n := \mathbb{E}^{\mathcal{D}}[\cdot \mid \mathcal{B}_{1/n}].$$

Proposition 4.7 (cf. Lemma 6.4 and Proposition 6.5, [2])**.** *There is a unique isometric isomorphism*

$$U_n : H_{1/n} \longrightarrow H_n = L^2(\widehat{M}_n, \widehat{m}_n)$$

with the following property: if

$$u(\eta) = F\big(\hat{f}_1^\star(\eta), \dots, \hat{f}_k^\star(\eta)\big), \qquad F \in C_b^\infty(\mathbb{R}^k),\ \hat{f}_j \in C_c^\infty(\mathbb{T}^d \times (1/n, 1]),$$

then

$$(U_n u)(s, x) = F\Big(\sum_{i=1}^n s_i \hat{f}_1(x_i, s_i), \dots, \sum_{i=1}^n s_i \hat{f}_k(x_i, s_i)\Big), \qquad (s, x) \in \widehat{M}_n.$$

If $J_n := U_n^{-1} : H_n \to H$, then J_n is an isometric embedding with

$$J_n(H_n) = H_{1/n} = \mathrm{Ran}(P_n).$$

Hence the finite-dimensional space H_n is canonically realized inside H through the orthogonal projection P_n.

Let $\hat{\mathfrak{Z}}_{1/n}^\infty$ denote the class of cylinder functions appearing in the statement.

Proposition 4.8 (Compatibility of finite- and infinite-dimensional gradients)**.** *Let*

$$\mathfrak{H} := L^2\big(P(\mathbb{T}^d) \times \mathbb{T}^d, \mathcal{D}; \mathbb{R}^d\big), \|F\|_{\mathfrak{H}}^2 = \int_{P(\mathbb{T}^d)} \int_{\mathbb{T}^d} |F(\eta, z)|^2\, \eta(\mathrm{d}z)\, \mathcal{D}(\mathrm{d}\eta)$$

For $V = (V_1, \dots, V_n) \in \mathfrak{H}_n$, define $K_n V \in \mathfrak{H}$ by

$$(K_n V)(\eta, z) := \sum_{i=1}^n \mathbf{1}_{\{x_i(\eta)\}}(z)\, V_i\big(\mathrm{pr}_n(s(\eta)), x_1(\eta), \dots, x_n(\eta)\big), \qquad (\eta, z) \in O \times \mathbb{T}^d,$$

and set $K_n V(\eta, z) = 0$ on $O^c \times \mathbb{T}^d$ or $s_1(\eta) < 1/n$. Here, $s(\eta), x_i(\eta)$ are defined by $\mathrm{em}^{-1}(\eta) = (s(\eta), x(\eta))$ with $s(\eta) = (s_1(\eta), s_2(\eta), \cdots)$ and $x(\eta) = (x_1(\eta), x_2(\eta), \dots)$, and O is the domain of em^{-1}. Then K_n is an isometric embedding of $\mathfrak{H}_n$ into $\mathfrak{H}$.

Moreover, for every $u \in H_n^{1,2}$,

$$J_n u \in H^{1,2}, \qquad D(J_n u) = K_n D^{(n)} u, \qquad \mathrm{CE}(J_n u) = \mathcal{E}_n(u). \tag{4.13}$$

In particular, if $u^{(n)} \in H_n^{1,2}$ and $J_n u^{(n)} \to u$ in $H^{1,2}$, then

$$K_n D^{(n)} u^{(n)} \to Du \qquad \text{in } \mathfrak{H}. \tag{4.14}$$

Proof. For any $V \in \mathfrak{H}_n$, we have

$$\begin{aligned}
\|K_n V\|_{\mathfrak{H}}^2 &= \int_{P(\mathbb{T}^d)} \int_{\mathbb{T}^d} |K_n V(\eta, z)|^2\, \eta(\mathrm{d}z)\, \mathcal{D}(\mathrm{d}\eta) \\
&= \int_{P(\mathbb{T}^d)} \sum_{i=1}^n s_i(\eta) \Big| V_i\big(\mathrm{pr}_n(s(\eta)), x_1(\eta), \dots, x_n(\eta)\big)\Big|^2\, \mathcal{D}(\mathrm{d}\eta) \\
&= \int_{\widehat{M}_n} \sum_{i=1}^n r_i |V_i(r, x)|^2\, \widehat{m}_n(\mathrm{d}r, \mathrm{d}x) = \|V\|_{\mathfrak{H}_n}^2.
\end{aligned}$$

Let $\hat{\mathfrak{Z}}_{1/n}^\infty$ be the cylinder class introduced in Proposition 4.7 and fix $v \in \hat{\mathfrak{Z}}_{1/n}^\infty$, say

$$v(\eta) = F\big(\hat{f}_1^\star(\eta), \dots, \hat{f}_k^\star(\eta)\big), \qquad F \in C_b^\infty(\mathbb{R}^k), \hat{f}_j \in C_c^\infty(\mathbb{T}^d \times (1/n, 1]).$$

Then $U_n v \in C^1(\widehat{M}_n)$ and

$$(U_n v)(r, x) = F\Big(\sum_{i=1}^n r_i \hat{f}_1(x_i, r_i), \dots, \sum_{i=1}^n r_i \hat{f}_k(x_i, r_i)\Big).$$

A direct differentiation in the torus variables yields

$$D_i^{(n)}(U_n v)(r,x) = \sum_{j=1}^{k} \partial_j F(\cdots)\, \nabla_x \hat{f}_j(x_i, r_i), \qquad i = 1, \dots, n.$$

Therefore,

$$K_n D^{(n)}(U_n, v)(\eta, z) = \sum_{i=1}^{n} \mathbf{1}_{\{x_i(\eta)\}}(z) \sum_{j=1}^{k} \partial_j F(\cdots)\, \nabla_x \hat{f}_j(x_i(\eta), s_i(\eta)).$$

which equals

$$\sum_{j=1}^{k} \partial_j F(\cdots)\, \nabla_x \hat{f}_j(x_i, s_i(\eta))$$

if $z = x_i(\eta)$ for some i and equals 0 otherwise. This is precisely the gradient formula for cylinder functions in the infinite-dimensional setting, as given by [1, Lemma 2.8]:

$$Dv(\eta, z) = \sum_{j=1}^{k} \partial_j F(\cdots)\, \nabla_x \hat{f}_j\big(z, \eta(\{z\})\big).$$

Hence $Dv = K_n D^{(n)}(U_n v)$ and in particular $D(J_n U_n v) = K_n D^{(n)}(U_n v)$ because $J_n = U_n^{-1}$. Moreover,

$$\mathrm{CE}(J_n U_n v) = \|Dv\|_{\mathfrak{H}}^2 = \|K_n D^{(n)}(U_n v)\|_{\mathfrak{H}}^2 = \|D^{(n)}(U_n v)\|_{\mathfrak{H}_n}^2 = \mathcal{E}_n(U_n v).$$

By Lemma 6.4 of [2] together with the density argument at the end of its proof, $U_n \hat{\mathfrak{Z}}_{1/n}^{\infty}$ is dense in $H_n^{1,2}$. Let now $u \in H_n^{1,2}$ and choose $u_k \in U_n \hat{\mathfrak{Z}}_{1/n}^{\infty}$ with $u_k \to u$ in $H_n^{1,2}$. Set $v_k \coloneqq J_n u_k$. Then

$$\|v_k - v_\ell\|_{H^{1,2}}^2 = \|u_k - u_\ell\|_{H_n}^2 + \mathrm{CE}(v_k - v_\ell) = \|u_k - u_\ell\|_{H_n}^2 + \mathcal{E}_n(u_k - u_\ell) = \|u_k - u_\ell\|_{H_n^{1,2}}^2.$$

so (v_k) is Cauchy in $H^{1,2}$. Hence by definition of closedness of the Dirichlet form, $v_k \to v$ for some $v \in H^{1,2}$ in $H^{1,2}$ and H. Since J_n is an isometry on H_n, we have $\|v_k - Ju\|_H = \|Ju_k - Ju\|_{H_n} = \|u_k - u\|_{H_n} \to 0$, so by uniqueness of limits we have $v = J_n u$. Passing to the limit in

$$Dv_k = K_n D^{(n)} u_k$$

and using that both D and $D^{(n)}$ are closed while K_n is an isometry, we obtain (4.13). Finally, (4.14) is immediate: if $J_n u^{(n)} \to u$ in $H^{1,2}$, then

$$\|K_n D^{(n)} u^{(n)} - Du\|_{\mathfrak{H}} = \|D(J_n u^{(n)}) - Du\|_{\mathfrak{H}} \to 0.$$

$\square$

From now on, whenever we compare H_n with H, we do so through the canonical embedding $J_n : H_n \to H_{1/n} \subset H$. That is, we identify the elements of H_n with their images under J_n in $H_{1/n}$. Moreover, we will always take $\Phi_n = P_n$ in the definition of $\mathcal{H}$-convergence. With this convention, the finite-dimensional space H_n is canonically realized as a closed subspace of H through the orthogonal projection P_n. We will now turn to the main result of this section, which is the convergence of $\tilde{P}_{s,t}^{(n)}$ to $\tilde{P}_{s,t}$ in the sense of $\mathcal{H}$-strong convergence. We remark that $P_{s,t}^{(n)}$ and $\tilde{P}_{s,t}$ involve products of functions, so we will use the following elementary lemma.

Lemma 4.9. *Let H_n, H as above and let $a_n \in H_n, a \in H, b_n \in H_n, b \in H$. Assume that*

- *$a_n \to a, b_n \to b$ in the sense of $\mathcal{H}$-strong convergence, and*
- *There exists a constant $C > 0$ so that $\sup_n \|a_n\|_\infty \le C$ and $\|a\|_\infty \le C$.*

Then $a_n b_n \to ab$ in the sense of $\mathcal{H}$-strong convergence.

Proof. Because $\mathcal{H}$-strong convergence is defined through the maps $\Phi_n = P_n$, Lemma A.10 of [2] shows that $\mathcal{H}$-strong convergence of a_n to a is equivalent to ordinary strong convergence of a_n to a in $H = L^2(P(\mathbb{T}^d), \mathcal{D})$, and similarly for b_n. It is therefore enough to prove that

$$\|a_n b_n - ab\|_{L^2(\mathcal{D})} \xrightarrow[n\to\infty]{} 0.$$

Write

$$a_n b_n - ab = a_n(b_n - b) + (a_n - a)b.$$

For the first term we simply use the uniform L^∞ bound on a_n:

$$\|a_n(b_n - b)\|_{L^2(\mathcal{D})} \le C\,\|b_n - b\|_{L^2(\mathcal{D})} \xrightarrow[n\to\infty]{} 0.$$

For the second term, fix $M > 0$ and let

$$b^M \coloneqq (-M) \vee (b \wedge M)$$

be the usual truncation of b. Then $\|b - b^M\|_{L^2(\mathcal{D})} \to 0$ as $M \to \infty$, and since $|a_n - a| \le 2C$ almost everywhere we have

$$\begin{aligned}\|(a_n - a)b\|_{L^2(\mathcal{D})} &\le \|(a_n - a)b^M\|_{L^2(\mathcal{D})} + \|(a_n - a)(b - b^M)\|_{L^2(\mathcal{D})}\\ &\le M\,\|a_n - a\|_{L^2(\mathcal{D})} + 2C\,\|b - b^M\|_{L^2(\mathcal{D})}.\end{aligned}$$

First let $n \to \infty$ and use the strong convergence $a_n \to a$ in H; then let $M \to \infty$. This shows that $\|(a_n - a)b\|_{L^2(\mathcal{D})} \to 0$.

Combining the two estimates yields $\|a_n b_n - ab\|_{L^2(\mathcal{D})} \to 0$, which is precisely the desired $\mathcal{H}$-strong convergence of $a_n b_n$ to ab. □

Theorem 4.10. *$\tilde{P}^{(n)}_{s,t} : H_n \to H_n$ converges $\mathcal{H}$-strongly to $\tilde{P}_{s,t} : H \to H$.*

Proof. By Proposition 4.7, after transport by J_n the operators $P_n = \mathbb{E}^{\mathcal{D}}[\cdot \mid \mathcal{B}_{1/n}]$ are orthogonal projections on H with range $J_n(H_n) = H_{1/n}$, and $P_n \to \mathrm{Id}_H$ strongly as $n \to \infty$. Hence, by [2, Lemma A.10], $\mathcal{H}$-strong convergence is the same as strong convergence in H. We will use this repeatedly below. We follow a similar strategy as Proposition 6.5 in [2]. Let $g \in L^\infty(P(\mathbb{T}^d), \mathcal{D})$ be a function with $g \ge c > 0$ for some constant $c > 0$, and define its finite-dimensional approximation by

$$g_n \coloneqq P_n g \in H_n. \tag{4.15}$$

Under this definition, we show:

(1) That $g_n = P_n g$ converges to g in H as $n \to \infty$.
(2) That $P^{(n)} \to P$ in the sense of $\mathcal{H}$-strong convergence.

Once we have these two points, we have that $u^{(n)}_t = P^{(n)}_{T-t} g_n$, converges to $u_t = P_{T-t} g$ in H for each $t \in [0, T]$ by definition of the $\mathcal{H}$-strong convergence of operators. Regarding the first point, as discussed in the proof of [2, Proposition 6.5], P_n converges strongly to Id, so $g_n = P_n g$ converges to g in H as $n \to \infty$. The second point follows from the fact that the corresponding Dirchlet forms converge in the sense of Mosco convergence, which is shown in [2, Proposition 6.5], and the Kuwae-Shioya theorem [2, Theorem A.11], which states the equivalence between Mosco convergence of Dirichlet forms and $\mathcal{H}$-strong convergence of the associated semigroups.

Next, we will show that, for any $f_n \in H_n$ $\mathcal{H}$-strongly converging to $f \in H$, $P^{(n)}_{t-s}(u^{(n)}_t f_n)$ converges to $P_{t-s}(u_t f)$ in H as $n \to \infty$. First, we will prove that, for any $f_n \in H_n$, $u^{(n)}_t f_n$ converges to $u_t f$ in the sense of $\mathcal{H}$-strong convergence. This follows from Lemma 4.9 and the fact that $u^{(n)}_t$ is uniformly bounded by $\|g\|_{L^\infty}$ for each t. Then, by the $\mathcal{H}$-strong convergence of $P^{(n)}_{t-s}$ to P_{t-s}, we have that $P^{(n)}_{t-s}(u^{(n)}_t f_n)$ converges to $P_{t-s}(u_t f)$ in H as $n \to \infty$.

Finally, we show that $\frac{1}{u_s^{(n)}}$ converges to $\frac{1}{u_s}$ in the sense of $\mathcal{H}$-strong convergence. We first note that $g_n = P_n g$ is bounded below by c since $g \geq c$ and P_n is a positive operator. By again the positivity, we have $u_t^{(n)} = P_{T-t}^{(n)} g_n \geq c$ for each $t \in [0,T]$. Thus, $\frac{1}{u_s^{(n)}}$ is well-defined and bounded by $1/c$. Now we note that a function $\phi : \mathbb{R} \to \mathbb{R}$ defined by $\phi(x) = 1/x$ is $1/c^2$-Lipschitz on $[c, \infty)$, so after transport to H we have

$$\left\| \frac{1}{u_s^{(n)}} - \frac{1}{u_s} \right\|_H = \|\phi(u_s^{(n)}) - \phi(u_s)\|_H \leq \frac{1}{c^2} \|u_s^{(n)} - u_s\|_H \to 0 \tag{4.16}$$

as $n \to \infty$.

We will now combine the above results to show that $\tilde{P}_{s,t}^{(n)} f$ converges to $\tilde{P}_{s,t} f$ in the sense of $\mathcal{H}$-strong convergence. This follows from the fact that $\frac{1}{u_s^{(n)}}$ and $\frac{1}{u_s}$ are uniformly bounded by $1/c$ for each n and Lemma 4.9 applied to the product of $\frac{1}{u_s^{(n)}}$ and $P_{t-s}^{(n)}(u_t^{(n)} f_n)$. □

5. Eulerian Approximation

We will now consider the Eulerian approximation of the diffusion. In the previous Lagrangian approach, the support points x_i of the Dirac measures moved and the mass vector s was frozen. In the Eulerian approximation, we will instead keep the support points fixed and let the mass vector evolve. To this end, we will set up a finite dimensional Dirichlet form and prove the Mosco convergence of the Eulerian Dirichlet forms to the infinite-dimensional one. We will first consider the free diffusion case and then consider the h-transformed dynamics.

5.1. **Basic definitions and notations.**

Grids. We will first define the grid notation to be used in the Eulerian approximation. We continue using the d-dimensional torus $\mathbb{T}^d$ as the underlying space and denote by m the normalized Lebesgue measure on $\mathbb{T}^d$. On the space of probability measures $P(\mathbb{T}^d)$, we will use the Wasserstein distance W_2 induced by the Euclidean distance on $\mathbb{T}^d$. For $n \in \mathbb{N}$, we set

$$h = 1/n \tag{5.1}$$

to be the mesh size of the approximation. We will denote by $\mathcal{I}^{[n]}$ the grid index set

$$\mathcal{I}^{[n]} = \{(i_1, \ldots, i_d) : i_j = 0, 1, \ldots, n-1\}, \tag{5.2}$$

and for $i = (i_1, \ldots, i_d) \in \mathcal{I}^{[n]}$ we define the grid cell

$$C_i^{[n]} = \prod_{j=1}^{d} \left[\frac{i_j}{n}, \frac{i_j + 1}{n} \right) \subset \mathbb{T}^d, \tag{5.3}$$

and its center

$$x_i^{[n]} = \left(\frac{i_1 + 1/2}{n}, \ldots, \frac{i_d + 1/2}{n} \right) \in \mathbb{T}^d. \tag{5.4}$$

We denote the set of all centers by $x^{[n]}$. For $l \in \{1, \cdots d\}$, let e_l be the l-th standard basis vector in $\mathbb{R}^d$. In the following, we will often write $i \pm e_l$ for the index obtained by adding/subtracting 1 to the l-th component of i, and we will identify $i \pm e_l$ with its representative in $\mathcal{I}^{[n]}$ modulo n.

The mass simplex and the reference measure. We will now consider the state space of the Eulerian approximation, i.e., the space of mass vectors. We will denote by $\Sigma^{[n]}$ the simplex of mass vectors

$$\Sigma^{[n]} = \left\{ s = (s_i)_{i\in\mathcal{I}^{[n]}} : s_i \geq 0, \sum_{i\in\mathcal{I}^{[n]}} s_i = 1 \right\}. \tag{5.5}$$

We also consider its relative interior

$$\Sigma^{\circ,[n]} = \left\{ s = (s_i)_{i\in\mathcal{I}^{[n]}} : s_i > 0, \sum_{i\in\mathcal{I}^{[n]}} s_i = 1 \right\}. \tag{5.6}$$

Given a mass vector $s \in \Sigma^{[n]}$, we consider the map $\iota^{[n]} : \Sigma^{[n]} \to \mathbb{P}(\mathbb{T}^d)$ that associates to s the probability measure

$$\iota^{[n]}(s) = \mathrm{em}(s, x^{[n]}) = \sum_{i\in\mathcal{I}^{[n]}} s_i \delta_{x_i^{[n]}}. \tag{5.7}$$

Conversely, we consider the map $\pi^{[n]} : \mathbb{P}(\mathbb{T}^d) \to \Sigma^{[n]}$ that associates to $\mu \in \mathbb{P}(\mathbb{T}^d)$ the mass vector

$$\pi^{[n]}(\mu) = \left(\mu(C_i^{[n]})\right)_{i\in\mathcal{I}^{[n]}}. \tag{5.8}$$

We then define the reference measure $\mathcal{D}^{[n]}$ on $\Sigma^{[n]}$ by the pushforward of the Dirichlet–Ferguson measure $\mathcal{D}$ under $\pi^{[n]}$:

$$\mathcal{D}^{[n]} = \pi^{[n]}_{\#}\mathcal{D}. \tag{5.9}$$

The defining property of the Dirichlet–Ferguson measure implies that $\mathcal{D}^{[n]}$ is exactly the Dirichlet distribution on $\Sigma^{[n]}$ with parameter $\alpha_i^{[n]} = \beta m(C_i^{[n]}) = \beta/n^d$ for each $i \in \mathcal{I}^{[n]}$. In particular, $\mathcal{D}^{[n]}$ has a smooth density

$$\rho^{[n]}(s) = \frac{\Gamma(\beta)}{\Gamma(\beta/n^d)^{n^d}} \prod_{i\in\mathcal{I}^{[n]}} s_i^{\beta/n^d-1} \tag{5.10}$$

with respect to the Lebesgue measure on $\Sigma^{[n]}$ and Γ is the Gamma function.

Edge calculus. For each grid point $(\frac{i_1}{n}, \dots, \frac{i_d}{n})$ corresponding to a grid index $i = (i_1, \cdots, i_d)$, we consider the d oriented edges in the direction of e_l, $l = 1, \cdots, d$, which we represent by (i,l). The edge (i,l) the tail index i to the head index $i+e_l$. On each edge (i,l), we consider the edge derivative operator $\partial_{i,l}^{[n]} F$ for a smooth function $F : \Sigma^{[n]} \to \mathbb{R}$:

$$\partial_{i,l}^{[n]} F(s) = \frac{\partial_{s_j} F(s) - \partial_{s_i} F(s)}{h}, \qquad j = i + e_l. \tag{5.11}$$

We note that a function $F : \Sigma^{[n]} \to \mathbb{R}$ is smooth if it can be extended to a smooth function on an open neighborhood of $\Sigma^{[n]}$ in $\mathbb{R}^{n^d}$. We remark that the edge derivative $\partial_{i,l}^{[n]} F$ does not depend on the choice of extension since $\partial_{i,l}^{[n]} F(s)$ is a directional derivative in the direction of $e_j - e_i$ which is tangent to $\Sigma^{[n]}$.

The free Eulerian Dirichlet form. To define the Dirichlet form, consider the L^2 space

$$H^{[n]} = L^2(\Sigma^{[n]}, \mathcal{D}^{[n]}). \tag{5.12}$$

For a smooth function $F, G : \Sigma^{[n]} \to \mathbb{R}$, we define

$$\mathcal{E}^{[n]}(F,G) = \sum_{i \in \mathcal{I}^{[n]}} \sum_{l=1}^{d} \int_{\Sigma^{[n]}} \theta_{i,l}(s) \partial_{i,l}^{[n]} F(s) \partial_{i,l}^{[n]} G(s) \mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.13}$$

Here, $\theta_{i,l}(s)$ is a function that defines the amount of mass on the edge i, l that we will specify later. Our goal from now on is to choose an appropriate $\theta_{i,l}$ so that the associated Dirichlet form $\mathcal{E}^{[n]}$ converges to the infinite-dimensional Dirichlet form $\mathcal{E}$ in the sense of generalized Mosco convergence.

5.2. **Failure of the arithmetic mean choice of $\theta_{i,l}$.** We first demonstrate that a naive choice

$$\theta_{i,l}(s) = \frac{s_i + s_j}{2}, \qquad j = i + e_l, \tag{5.14}$$

does not yield the desired convergence. We begin by recalling the definition of the generalized Mosco convergence of quadratic forms.

Definition 5.1 (Generalized Mosco convergence of quadratic forms. [2])**.** Let $(Q_n, \mathcal{D}(Q_n))$ be a sequence of quadratic forms on Hilbert spaces H_n and $(Q, \mathcal{D}(Q))$ be a quadratic form on another Hilbert space H. We say that Q_n converges to Q in the sense of generalized Mosco convergence if the following conditions hold:

(1) H_n converges to H in the sense of $\mathcal{H}$-convergence.
(2) If $u_n \in H_n$ converges $\mathcal{H}$-weakly to $u \in H$, then $\liminf_{n\to\infty} Q_n(u_n) \geq Q(u)$.
(3) For every $u \in H$, there exists a sequence $u_n \in H_n$ converging $\mathcal{H}$-strongly to u such that $\limsup_{n\to\infty} Q_n(u_n) = Q(u)$.

We will now show that the liminf condition fails for the naive choice of $\theta_{i,l}$.

Proposition 5.2. *Let $d = 1$ and suppose n is even. For any smooth nonconstant function $f : \mathbb{T}^1 \to \mathbb{R}$, consider a function $F : P(\mathbb{T}^1) \to \mathbb{R}$ defined by*

$$F(\mu) = \int_{\mathbb{T}^1} f(x) \mu(\mathrm{d}x). \tag{5.15}$$

Consider the projection $\Lambda_n : L^2(P(\mathbb{T}^1), \mathcal{D}) \to H^{[n]}$ defined by Then there exists $F_n : \Sigma^{[n]} \to \mathbb{R}$ converging to F in the sense of $\mathcal{H}$-strong convergence, but

$$\lim_{n\to\infty} \mathcal{E}^{[n]}(F_n, F_n) < \mathcal{E}(F, F). \tag{5.16}$$

Here, $\mathcal{E}$ is the infinite-dimensional Dirichlet form defined in (2.12).

Proof. The idea of the proof is as follows: since $\mathcal{D}^{[n]}$ is Dirichlet distribution with parameter β/n, this parameter shrinks as $n \to \infty$. This leads to a sample s from $\mathcal{D}^{[n]}$ being close to the vertices of the simplex $\Sigma^{[n]}$. In particular, the edge weight $\theta_{i,l}(s) = (s_i + s_j)/2$ will be close to 0 for most of the edges, which leads to a vanishing contribution to the Dirichlet form from those edges. This results in underestimating the energy of the limit function F.

Step 1: Construction of the recovery sequence. To make the above idea precise, we first construct a recovery sequence F_n for F. We have

$$\Lambda_n F(s) = \sum_{i=0}^{n-1} s_i f_i = s_0 f_0 + s_1 f_1 + \cdots + s_{n-1} f_{n-1} \tag{5.17}$$

$$= (s_0 f_0 + s_1 f_1) + (s_2 f_2 + s_3 f_3) + \cdots + (s_{n-2} f_{n-2} + s_{n-1} f_{n-1}) \tag{5.18}$$

$$= \sum_{k=0}^{n/2-1} (s_{2k} f_{2k} + s_{2k+1} f_{2k+1}) \tag{5.19}$$

$$= \sum_{k=0}^{n/2-1} (s_{2k} + s_{2k+1}) \left(f_{2k} \frac{s_{2k}}{s_{2k} + s_{2k+1}} + f_{2k+1} \left(1 - \frac{s_{2k}}{s_{2k} + s_{2k+1}} \right) \right) \tag{5.20}$$

$$= \sum_{k=0}^{n/2-1} (s_{2k} + s_{2k+1}) \left(f_{2k} r_k + f_{2k+1}(1 - r_k) \right) \tag{5.21}$$

Here, we have defined $r_k = s_{2k}/(s_{2k} + s_{2k+1})$. We now define $F_n : \Sigma^{[n]} \to \mathbb{R}$ by

$$F_n(s) = \sum_{k=0}^{n/2-1} (s_{2k} + s_{2k+1}) \left(f_{2k} \chi(r_k) + f_{2k+1}(1 - \chi(r_k)) \right). \tag{5.22}$$

The formula is understood on the set where $s_{2k} + s_{2k+1} > 0$ for all k; the complement is $\mathcal{D}^{[n]}$-negligible. Here, $\chi : [0,1] \to [0,1]$ is a smooth monotone function so that $\chi(0) = 0, \chi(1) = 1, \chi'(0) = \chi'(1) = \epsilon$ for some $\epsilon \in (1/3, 1)$. For concreteness, we take

$$\chi(x) = \frac{2}{3}x + x^2 - \frac{2}{3}x^3 \tag{5.23}$$

which satisfies the above properties with $\epsilon = 2/3$. The transformation here is done to expose the contribution from the edge $(2k, 2k+1)$ to the integral, which is $f_{2k} r_k + f_{2k+1}(1 - r_k)$. We then compose by χ which has derivative strictly smaller than one at the boundary points $0, 1$ where the mass concentrates as $n \to \infty$, aiming to make the energy smaller than the energy of the limit function. We first prove that F_n converges to F in the sense of $\mathcal{H}$-strong convergence. By smoothness, $\Lambda_n F$ converges to F in H as $n \to \infty$. Moreover, we have

$$F_n - \Lambda_n F = \sum_{k=0}^{n/2-1} (s_{2k} + s_{2k+1}) \left(f_{2k} - f_{2k+1} \right) \left(\chi(r_k) - r_k \right). \tag{5.24}$$

By using the fact that $\sum_{k=0}^{n/2-1} (s_{2k} + s_{2k+1}) = 1$, that $\chi(x) - x$ is uniformly bounded, and that $|f_{2k} - f_{2k+1}| \leq \frac{C}{n}$ for some constant C depending on f, we have $\|F_n - \Lambda_n F\|_{H^{[n]}} \leq \frac{C'}{n}$ for some constant C' that depends on f and χ. Hence, F_n converges to F in H as $n \to \infty$, which is the desired $\mathcal{H}$-strong convergence.

Step 2: Computation of the energy. We now compute the energy $\mathcal{E}^{[n]}(F_n, F_n)$. The discrete energy is

$$\mathcal{E}^{[n]}(F_n, F_n) = \sum_{i=0}^{n-1} \int_{\Sigma^{[n]}} \frac{s_i + s_{i+1}}{2} \left(\frac{\partial_{s_{i+1}} F_n(s) - \partial_{s_i} F_n(s)}{h} \right)^2 \mathcal{D}^{[n]}(\mathrm{d}s),$$

where all indices are understood modulo n. We also understand the pair index $k+1$ modulo $n/2$ below, so that $r_{n/2} = r_0$. The sum can be broken down into two parts depending on if i is even or odd:

$$\mathcal{E}^{[n]}(F_n, F_n) = \sum_{k=0}^{n/2-1} \int_{\Sigma^{[n]}} \frac{s_{2k}+s_{2k+1}}{2} \left(\frac{\partial_{s_{2k+1}} F_n(s) - \partial_{s_{2k}} F_n(s)}{h}\right)^2 \mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.25}$$

$$+ \sum_{k=0}^{n/2-1} \int_{\Sigma^{[n]}} \frac{s_{2k+1}+s_{2k+2}}{2} \left(\frac{\partial_{s_{2k+2}} F_n(s) - \partial_{s_{2k+1}} F_n(s)}{h}\right)^2 \mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.26}$$

We refer to the first sum as the even sum and the second sum as the odd sum. We first calculate the even sum.

Step 2.1: Calculation of the even sum. We first note that, by quotient rule, we have

$$\partial_{s_{2k}} r_k = \frac{s_{2k+1}}{(s_{2k}+s_{2k+1})^2} = \frac{1-r_k}{s_{2k}+s_{2k+1}}, \qquad \partial_{s_{2k+1}} r_k = -\frac{s_{2k}}{(s_{2k}+s_{2k+1})^2} = -\frac{r_k}{s_{2k}+s_{2k+1}}. \tag{5.27}$$

Therefore, we have

$$\partial_{s_{2k}} F_n(s) = \partial_{s_{2k}} (s_{2k}+s_{2k+1})(f_{2k}\chi(r_k) + f_{2k+1}(1-\chi(r_k))) \tag{5.28}$$

$$= (f_{2k}\chi(r_k) + f_{2k+1}(1-\chi(r_k))) + (s_{2k}+s_{2k+1}) \left(f_{2k}\chi'(r_k)\frac{1-r_k}{s_{2k}+s_{2k+1}} - f_{2k+1}\chi'(r_k)\frac{1-r_k}{s_{2k}+s_{2k+1}}\right) \tag{5.29}$$

$$= f_{2k}\chi(r_k) + f_{2k+1}(1-\chi(r_k)) + (f_{2k} - f_{2k+1})\chi'(r_k)(1-r_k) \tag{5.30}$$

On the other hand, we have

$$\partial_{s_{2k+1}} F_n(s) = \partial_{s_{2k+1}} (s_{2k}+s_{2k+1})(f_{2k}\chi(r_k) + f_{2k+1}(1-\chi(r_k))) \tag{5.31}$$

$$= (f_{2k}\chi(r_k) + f_{2k+1}(1-\chi(r_k))) + (s_{2k}+s_{2k+1}) \left(f_{2k}\chi'(r_k)\frac{-r_k}{s_{2k}+s_{2k+1}} - f_{2k+1}\chi'(r_k)\frac{-r_k}{s_{2k}+s_{2k+1}}\right) \tag{5.32}$$

$$= f_{2k}\chi(r_k) + f_{2k+1}(1-\chi(r_k)) - (f_{2k} - f_{2k+1})\chi'(r_k) r_k \tag{5.33}$$

Therefore, we obtain

$$\partial_{s_{2k+1}} F_n(s) - \partial_{s_{2k}} F_n(s) = -(f_{2k} - f_{2k+1})\chi'(r_k). \tag{5.34}$$

Therefore, the even sum is equal to

$$\sum_{k=0}^{n/2-1} \mathbb{E}_{\mathcal{D}^{[n]}} \left[\frac{s_{2k}+s_{2k+1}}{2} \left(\frac{\partial_{s_{2k+1}} F_n(s) - \partial_{s_{2k}} F_n(s)}{h}\right)^2\right] \tag{5.35}$$

$$= \sum_{k=0}^{n/2-1} \mathbb{E}_{\mathcal{D}^{[n]}} \left[\frac{s_{2k}+s_{2k+1}}{2} \left(\frac{(f_{2k} - f_{2k+1})\chi'(r_k)}{h}\right)^2\right] \tag{5.36}$$

$$= \frac{1}{2} \sum_{k=0}^{n/2-1} \left(\frac{(f_{2k} - f_{2k+1})}{h}\right)^2 \mathbb{E}_{\mathcal{D}^{[n]}} \left[(s_{2k}+s_{2k+1})\chi'(r_k)^2\right] \tag{5.37}$$

By the standard property of the Dirichlet distribution, we have that r_k is independent of $s_{2k}+s_{2k+1}$ and r_k follows a Beta distribution with parameters β/n and β/n. Moreover, the Beta distribution with parameters β/n and β/n converges weakly to the Bernoulli distribution that assigns mass 1/2 to 0 and mass 1/2 to 1 as $n \to \infty$. Therefore, we have $\mathbb{E}_{\mathcal{D}[n]}[\chi'(r_k)^2] \to \frac{1}{2}(\chi'(0)^2 + \chi'(1)^2) = \frac{4}{9}$ as $n \to \infty$. On the other hand, we have $\mathbb{E}_{\mathcal{D}[n]}[s_{2k} + s_{2k+1}] = 2/n = 2h$. Therefore, we have

$$\sum_{k=0}^{n/2-1} \mathbb{E}_{\mathcal{D}[n]} \left[\frac{s_{2k} + s_{2k+1}}{2} \left(\frac{\partial_{s_{2k+1}} F_n(s) - \partial_{s_{2k}} F_n(s)}{h} \right)^2 \right] \tag{5.38}$$

$$= \frac{1}{2} \sum_{k=0}^{n/2-1} \left(\frac{(f_{2k} - f_{2k+1})}{h} \right)^2 2h \cdot \frac{4}{9} + o(1) \tag{5.39}$$

$$= \frac{4}{9} \sum_{k=0}^{n/2-1} \left(\frac{f_{2k} - f_{2k+1}}{h} \right)^2 h + o(1) \tag{5.40}$$

$$\to \frac{2}{9} \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x \quad \text{as } n \to \infty. \tag{5.41}$$

Here the factor 1/2 in the last line comes from the fact that the sum only runs over every second edge of the grid.

Step 2.2: Calculation of the odd sum. We now calculate the odd sum. We first note that, by shifting the indices in the previous calculation, we have

$$\partial_{s_{2k+2}} F_n(s) = f_{2k+2}\chi(r_{k+1}) + f_{2k+3}(1 - \chi(r_{k+1})) + (f_{2k+2} - f_{2k+3})\chi'(r_{k+1})(1 - r_{k+1}) \tag{5.42}$$

$$= f_{2k+2}\chi(r_{k+1}) + f_{2k+3} - f_{2k+3}\chi(r_{k+1}) + (f_{2k+2} - f_{2k+3})\chi'(r_{k+1})(1 - r_{k+1}) \tag{5.43}$$

$$= (f_{2k+2} - f_{2k+3})\chi'(r_k) + f_{2k+3} + (f_{2k+2} - f_{2k+3})\chi'(r_{k+1})(1 - r_{k+1}) \tag{5.44}$$

$$= f_{2k+3} + (f_{2k+2} - f_{2k+3}) \left(\chi(r_{k+1}) + (1 - r_{k+1})\chi'(r_{k+1}) \right) \tag{5.45}$$

$$\partial_{s_{2k+1}} F_n(s) = f_{2k}\chi(r_k) + f_{2k+1}(1 - \chi(r_k)) - (f_{2k} - f_{2k+1})\chi'(r_k) r_k. \tag{5.46}$$

$$= f_{2k+1} + (f_{2k} - f_{2k+1}) \left(\chi(r_k) - r_k \chi'(r_k) \right) \tag{5.47}$$

Therefore,

$$\frac{\partial_{s_{2k+2}} F_n(s) - \partial_{s_{2k+1}} F_n(s)}{h} \tag{5.48}$$

$$= \frac{f_{2k+3} - f_{2k+1}}{h} + \frac{f_{2k+2} - f_{2k+3}}{h} \left(\chi(r_{k+1}) + (1 - r_{k+1})\chi'(r_{k+1}) \right) \tag{5.49}$$

$$- \frac{f_{2k} - f_{2k+1}}{h} \left(\chi(r_k) - r_k \chi'(r_k) \right). \tag{5.50}$$

By the same property of the Dirichlet distribution, r_k and r_{k+1} are independent Beta random variables with parameters β/n and β/n, and they are independent of the masses $s_{2k} + s_{2k+1}$ and $s_{2k+2} + s_{2k+3}$. Since

$$s_{2k+1} = (s_{2k} + s_{2k+1})(1 - r_k), \qquad s_{2k+2} = (s_{2k+2} + s_{2k+3}) r_{k+1},$$

We have

$$s_{2k+1} + s_{2k+2} = (s_{2k} + s_{2k+1})(1 - r_k) + (s_{2k+2} + s_{2k+3}) r_{k+1} \tag{5.51}$$

and the expectation of $s_{2k+1} + s_{2k+2}$ and $s_{2k+2} + s_{2k+3}$ are $2h$ Therefore, the odd sum is equal to

$$h \sum_{k=0}^{n/2-1} \mathbb{E}_{\mathcal{D}[n]} \left[(1 - r_k + r_{k+1}) \left(\frac{\partial_{s_{2k+2}} F_n(s) - \partial_{s_{2k+1}} F_n(s)}{h} \right)^2 \right]. \tag{5.52}$$

As $n \to \infty$, the pair (r_k, r_{k+1}) converges weakly to two independent Bernoulli random variables. Moreover, by smoothness of f, the four possible limiting values of the derivative quotient above are

$$(r_k, r_{k+1}) = (0,0): \quad (2-\epsilon)f'(x), \tag{5.53}$$

$$(r_k, r_{k+1}) = (0,1): \quad f'(x), \tag{5.54}$$

$$(r_k, r_{k+1}) = (1,0): \quad (3-2\epsilon)f'(x), \tag{5.55}$$

$$(r_k, r_{k+1}) = (1,1): \quad (2-\epsilon)f'(x). \tag{5.56}$$

The error in this replacement is $o(1)$ uniformly in k by smoothness of f. We remark that $\frac{f_{2k+3}-f_{2k+1}}{h}$ converges to $2f'(x)$ since the grid points are two steps apart, which leads to the factor 2 appearing in several cases. The corresponding values of $1 - r_k + r_{k+1}$ are $1, 2, 0, 1$ respectively. Therefore, we have

$$\sum_{k=0}^{n/2-1} \int_{\Sigma^{[n]}} \frac{s_{2k+1}+s_{2k+2}}{2} \left(\frac{\partial_{s_{2k+2}} F_n(s) - \partial_{s_{2k+1}} F_n(s)}{h} \right)^2 \mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.57}$$

$$\to \frac{1}{2} \left(\frac{1}{4}(2-\epsilon)^2 + \frac{1}{4} \cdot 2 + \frac{1}{4} \cdot 0 \cdot (3-2\epsilon)^2 + \frac{1}{4}(2-\epsilon)^2 \right) \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x \tag{5.58}$$

$$= \left(\frac{1}{4}(2-\epsilon)^2 + \frac{1}{4} \right) \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x. \tag{5.59}$$

Here, the factor $1/2$ after taking the limit again comes from the fact that the sum only runs over every second edge of the grid. For our choice $\epsilon = 2/3$, this gives

$$\text{odd sum} \longrightarrow \frac{25}{36} \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x. \tag{5.60}$$

Combining this with the even sum, we obtain

$$\lim_{n\to\infty} \mathcal{E}^{[n]}(F_n, F_n) = \left(\frac{2}{9} + \frac{25}{36} \right) \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x = \frac{11}{12} \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x. \tag{5.61}$$

On the other hand, for the linear cylinder function $F(\mu) = \int f \, \mathrm{d}\mu$ we have $DF(\mu, x) = f'(x)$ by definition of the Wasserstein gradient for cylinder functions. Hence, using (2.12) and the fact that the mean measure of the Dirichlet–Ferguson measure is the base measure, i.e., the Lebesgue measure in our case, we have

$$\mathcal{E}(F,F) = \int_{P(\mathbb{T}^1)} \int_{\mathbb{T}^1} |f'(x)|^2 \mu(\mathrm{d}x) \mathcal{D}(\mathrm{d}\mu) = \int_{\mathbb{T}^1} |f'(x)|^2 \mathrm{d}x. \tag{5.62}$$

Since f is nonconstant, the last integral is strictly positive. Therefore

$$\lim_{n\to\infty} \mathcal{E}^{[n]}(F_n, F_n) = \frac{11}{12} \mathcal{E}(F,F) < \mathcal{E}(F,F). \tag{5.63}$$

Since F_n converges $\mathcal{H}$-strongly to F, it also converges $\mathcal{H}$-weakly to F. This contradicts the liminf inequality required in the Mosco convergence definition, and so the naive choice $\theta_{i,l}(s) = (s_i + s_j)/2$ cannot yield Mosco convergence. □

5.3. **The harmonic mean choice of $\theta_{i,l}$ and the consistency property.** We will now specify the choice of $\theta_{i,l}$ that yields the desired convergence. The issue of the arithimetic mean choice is that the edge weight $\theta_{i,l}(s)$ vanishes as $n \to \infty$ for most of the edges, which leads to underestimating the energy of the limit function. To fix this issue, we will choose $\theta_{i,l}(s)$ to be the harmonic mean of s_i and s_j, which is given by

$$\theta_{i,l}(s) = \frac{2}{\frac{1}{s_i} + \frac{1}{s_j}} = \frac{2 s_i s_j}{s_i + s_j}, \qquad j = i + e_l. \tag{5.64}$$

However, the bare harmonic choice does not yield the desired convergence either. The main reason is that for this choice, we do not have the consistency property

$$\lim_{n\to\infty} \mathcal{E}^{[n]}(\Lambda_n F, \Lambda_n F) = \mathcal{E}(F, F) \tag{5.65}$$

for all F in the algebra of cylinder functions, whereas the arithmetic mean choice does have this property, as we will show in Proposition 5.5. To fix this issue, we will find a correction factor to the harmonic mean choice so that

$$\mathbb{E}\left[C_n \cdot \frac{s_i s_j}{s_i + s_j} \middle| s_i + s_j = r\right] = \frac{r}{2} \tag{5.66}$$

for all $r \in (0, 1)$ so that the new choice of θ is equal to the arithmetic mean in expectation, and hence the consistency property is satisfied.

Proposition 5.3. *Suppose $s_i, s_j, i \neq j$ are components of a vector sampled from a Dirichlet distribution with constant parameter α. For any $r \in (0, 1)$, we have*

$$\mathbb{E}\left[\frac{s_i s_j}{s_i + s_j} \middle| s_i + s_j = r\right] = \frac{r}{2} \cdot \frac{\alpha}{2\alpha + 1}. \tag{5.67}$$

Proof. We note that if we set $X = \frac{s_i}{s_i+s_j}$, then X follows a Beta distribution with parameters α and α. Moreover, we have $\frac{s_i s_j}{s_i+s_j} = rX(1-X)$. Therefore, by using the density of the Beta distribution, we have the desired formula. $\square$

By this result, we can choose the correction factor to be $C_n = \frac{2\alpha+1}{\alpha}$ and the new choice of θ is given by

$$\theta_{i,l}(s) = \frac{2\alpha + 1}{\alpha} \cdot \frac{s_i s_j}{s_i + s_j}, \qquad j = i + e_l. \tag{5.68}$$

We will now show that this choice of θ yields the finite dimensional Dirichlet form that is consistent with the infinite-dimensional Dirichlet form. To this en,d we first prove the following lemma.

Lemma 5.4. *Let J be a finite index set, let $i, j \in J$ with $i \neq j$, and let $\psi : \mathbb{R}^J \to \mathbb{R}$ be bounded and measurable. Assume that there exists $L > 0$ such that for every $r \in (0, 1)$ and every choice of fixed coordinates $a = (a_k)_{k\in J\setminus\{i,j\}} \in \mathbb{R}^{J\setminus\{i,j\}}$, the function*

$$\Psi_{a,r}(x) := \psi(s^{a,r}(x)), \qquad x \in [0, 1],$$

where

$$s_k^{a,r}(x) = \begin{cases} rx, & k = i, \\ r(1-x), & k = j, \\ a_k, & k \notin \{i, j\}, \end{cases}$$

is L-Lipschitz. Then, if $s = (s_k)_{k\in J}$ is a random vector with Dirichlet distribution with constant parameter $\alpha > 0$, we have

$$\left|\mathbb{E}\left[\left(\frac{2\alpha + 1}{\alpha} \frac{s_i s_j}{s_i + s_j} - \frac{s_i + s_j}{2}\right) \psi(s)\right]\right| \leq L\mathbb{E}[s_i + s_j].$$

Proof. Let

$$X = \frac{s_i}{s_i + s_j}, \quad R = s_i + s_j, \quad \hat{S} = (s_k)_{k\in J\setminus\{i,j\}}. \tag{5.69}$$

Then the left-hand side of the desired inequality can be rewritten as, for $x_0 \in [0,1]$,

$$\left|\mathbb{E}\left[\left(\frac{2\alpha+1}{\alpha}RX(1-X)-\frac{R}{2}\right)\psi(s)\right]\right| \tag{5.70}$$

$$=\left|\mathbb{E}\left[R\left(\frac{2\alpha+1}{\alpha}X(1-X)-\frac{1}{2}\right)\psi(s)\right]\right| \tag{5.71}$$

$$=\left|\mathbb{E}\left[R\left(\frac{2\alpha+1}{\alpha}X(1-X)-\frac{1}{2}\right)(\psi(s)-\psi(s^{\hat{S},R}(x_0)))\right]\right| \tag{5.72}$$

$$\leq \mathbb{E}\left[R\left|\frac{2\alpha+1}{\alpha}X(1-X)-\frac{1}{2}\right|\cdot|\psi(s)-\psi(s^{\hat{S},R}(x_0))|\right] \tag{5.73}$$

$$\leq \mathbb{E}[R]\cdot\left(\mathbb{E}\left[\left|\frac{2\alpha+1}{\alpha}X(1-X)\right|\right]+\mathbb{E}\left[\frac{1}{2}\right]\right)L \tag{5.74}$$

$$=\mathbb{E}[R]\cdot\left(\mathbb{E}\left[\frac{2\alpha+1}{\alpha}X(1-X)\right]+\mathbb{E}\left[\frac{1}{2}\right]\right)L \tag{5.75}$$

$$=L\mathbb{E}[R]=L\mathbb{E}[s_i+s_j]. \tag{5.76}$$

Here, from the second step to the third step, we have used the fact that $\mathbb{E}[\frac{2\alpha+1}{\alpha}X(1-X)-\frac{1}{2}]=0$ by the choice of the correction factor. From the fourth step to the fifth step, we have used the fact that X and $R,\hat{S}$ are independent. $\square$

We will now show that the new choice of θ yields the desired consistency property. We first prove the following proposition, which is a direct consequence of Lemma 5.4. We remark that the consistency also holds for the arithmetic mean choice of θ, which is not contradictory to the failure of Mosco convergence for that choice, since the consistency only holds for the projection of the cylinder functions, which is a small subset of the domain of the Dirichlet form.

Proposition 5.5. *Suppose U,V are two cylinder functions on $P(\mathbb{T}^d)$. Then we have*

$$\lim_{n\to\infty}\|\Lambda_n U\|_{H^{[n]}}=\|U\|_H, \qquad \lim_{n\to\infty}\mathcal{E}^{[n]}(\Lambda_n U,\Lambda_n V)=\mathcal{E}(U,V). \tag{5.77}$$

Here, for a cylinder function U of the form

$$U(\mu)=F\left(\int f_1\,\mathrm{d}\mu,\ldots,\int f_m\,\mathrm{d}\mu\right), \tag{5.78}$$

with $F\in C_b^\infty(\mathbb{R}^m)$ and $f_1,\ldots,f_m\in C^\infty(\mathbb{T}^d)$, the projection $\Lambda_n : H\to H^{[n]}$ is the grid projection defined by

$$\Lambda_n U(s)=U(\iota^{[n]}(s))=F\left(\sum_{i\in\mathcal{I}^{[n]}}s_i f_{1,i},\ldots,\sum_{i\in\mathcal{I}^{[n]}}s_i f_{m,i}\right),\quad s_i=\mu(C_i^{[n]}),\quad f_{k,i}=f_k(x_i^{[n]}). \tag{5.79}$$

Proof. **Step 1. Norm convergence.**

We first prove the norm convergence. By definition of the norm, we have

$$\|\Lambda_n U\|^2_{H^{[n]}}=\int_{\Sigma^{[n]}}U(\iota^{[n]}(s))^2\mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.80}$$

Recalling that $\mathcal{D}^{[n]}=\pi_n\#\mathcal{D}$, we have

$$\int_{\Sigma^{[n]}}U(\iota^{[n]}(s))^2\mathcal{D}^{[n]}(\mathrm{d}s)=\int_{P(\mathbb{T}^d)}U(\iota^{[n]}\circ\pi_n(\mu))^2\mathcal{D}(\mathrm{d}\mu). \tag{5.81}$$

We will now show that $\iota^{[n]}\circ\pi_n(\mu)$ converges to μ in the weak topology for every μ as $n\to\infty$ Since U is bounded and continuous with respect to the weak topology, this will imply the desired norm convergence by dominated convergence theorem. Since the weak topology is metrized by the

Wasserstein 2-distance, it suffices to show that $W_2(\iota^{[n]} \circ \pi_n(\mu), \mu) \to 0$ as $n \to \infty$. The coupling $\gamma_n = (\mathrm{id}, \iota^{[n]} \circ \pi_n)\#\mu$ is a coupling between μ and $\iota^{[n]} \circ \pi_n(\mu)$, and we have

$$\int_{(\mathbb{T}^d)^2} d_{\mathbb{T}^d}(x,y)^2 \gamma_n(\mathrm{d}x, \mathrm{d}y) = \int_{P(\mathbb{T}^d)} \int_{\mathbb{T}^d} d_{\mathbb{T}^d}(x, \iota^{[n]} \circ \pi_n(x))^2 \mu(\mathrm{d}x) \mathcal{D}(\mathrm{d}\mu), \tag{5.82}$$

where $d_{\mathbb{T}^d}$ is the distance on the torus. Since $\iota^{[n]} \circ \pi_n(x)$ is the center of the grid cell containing x, we have $d_{\mathbb{T}^d}(x, \iota^{[n]} \circ \pi_n(x)) \leq \sqrt{d}/n$ for every x. Therefore, we have

$$\int_{(\mathbb{T}^d)^2} d_{\mathbb{T}^d}(x,y)^2 \gamma_n(\mathrm{d}x, \mathrm{d}y) \leq \frac{d}{n^2} \to 0 \quad \text{as } n \to \infty. \tag{5.83}$$

Therefore we have the norm convergence.

Step 2. Energy convergence. We now prove the energy convergence. The strategy is as follows: we first prove that the energy with θ as the harmonic mean with correction factor converges to the same limit as the energy with θ as the arithmetic mean, and then we prove that the energy with θ as the arithmetic mean converges to the desired limit.

Step 2.1. Convergence of the energy with the harmonic mean choice. We first write down the approximate energy explicitly:

$$\mathcal{E}^{[n]}(\Lambda_n U, \Lambda_n V) = \sum_{i \in \mathcal{I}^{[n]}} \sum_{l=1}^{d} \int_{\Sigma^{[n]}} \frac{2\alpha+1}{\alpha} \cdot \frac{s_i s_{i+e_l}}{s_i + s_{i+e_l}} \partial_{i,l}^{[n]} \Lambda_n U(s) \partial_{i,l}^{[n]} \Lambda_n V(s) \mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.84}$$

Here, $\alpha = \frac{\beta}{n^d}$ is the parameter of the Dirichlet distribution $\mathcal{D}^{[n]}$. We first fix an edge $(i,l), i \in \mathcal{I}^{[n]}, l \in \{1, \ldots, d\}$ and write $j = i + e_l$. We will show that the mapping

$$s \mapsto \psi(s) = \partial_{i,l}^{[n]} \Lambda_n U(s) \partial_{i,l}^{[n]} \Lambda_n V(s) \tag{5.85}$$

satisfies the assumptions of Lemma 5.4. To see this, we first note that $\Lambda_n U$ and $\Lambda_n V$ are of the form

$$\Lambda_n U(s) = F(A_f(s)), \quad \Lambda_n V(s) = G(A_g(s)), \quad A_f(s) = \left(\sum_{i \in \mathcal{I}^{[n]}} s_i f_{1,i}, \ldots, \sum_{i \in \mathcal{I}^{[n]}} s_i f_{m,i} \right) \tag{5.86}$$

where $F, G \in C_b^\infty(\mathbb{R}^m)$ and $f_{k,i} = f_k(x_i^{[n]}), g_{k,i} = g_k(x_i^{[n]})$ for some $f_1, \ldots, f_m, g_1, \ldots, g_m \in C^\infty(\mathbb{T}^d)$. Therefore, we have

$$\partial_{i,l}^{[n]} \Lambda_n U(s) \partial_{i,l}^{[n]} \Lambda_n V(s) = \left(\nabla F(A_f(s)) \cdot \frac{f_{*,j} - f_{*,i}}{h} \right) \left(\nabla G(A_g(s)) \cdot \frac{g_{*,j} - g_{*,i}}{h} \right) \tag{5.87}$$

Here, ∇F is the gradient of F in $\mathbb{R}^m$, $f_{*,i} = (f_{1,i}, \ldots, f_{m,i})$, and similarly for ∇G and $g_{*,i}$. Now consider the function $\Psi_{a,r}$ defined in the statement of Lemma 5.4. We will show that this function has a bounded derivative. We have that

$$\frac{d}{dx} A_f(s^{a,r}(x)) = \frac{d}{dx} \left(\sum_{k \in \mathcal{I}^{[n]}} s_k^{a,r}(x) f_{*,k} \right) = r(f_{*,i} - f_{*,j}) \tag{5.88}$$

and similarly for A_g, so we have

$$\frac{d}{dx} \Psi_{a,r}(x) = r \nabla^2 F(A_f(s^{a,r}(x)))(f_{*,i} - f_{*,j}) \left(\nabla G(A_g(s^{a,r}(x))) \cdot \frac{g_{*,j} - g_{*,i}}{h} \right) \tag{5.89}$$

$$+ r \left(\nabla F(A_f(s^{a,r}(x))) \cdot \frac{f_{*,j} - f_{*,i}}{h} \right) \nabla^2 G(A_g(s^{a,r}(x)))(g_{*,i} - g_{*,j}). \tag{5.90}$$

We remark that by smoothness of F and G and the fact that domain of x is compact, the derivatives of F and G are bounded. Moreover, since f_k and g_k are smooth, the difference quotients $\frac{f_{*,j} - f_{*,i}}{h}$

and $\frac{g_{*,j}-g_{*,i}}{h}$ are also bounded uniformly in n. Finally, $f_{*,j}-f_{*,i}$ and $g_{*,j}-g_{*,i}$ are also bounded by $1/n$ times the Lipschitz constant of f_k and g_k. Therefore, we have that $\frac{d}{dx}\Psi_{a,r}(x)$ is bounded uniformly in n, which implies that $\Psi_{a,r}$ is $O(1/n)$-Lipschitz. Therefore, Lemma 5.4 applies, and for some constant L independent of n, we have

$$\left|\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\int_{\Sigma^{[n]}}\int_{\Sigma^{[n]}}\left(\frac{2\alpha+1}{\alpha}\cdot\frac{s_is_j}{s_i+s_j}-\frac{s_i+s_j}{2}\right)\partial_{i,l}^{[n]}\Lambda_nU(s)\partial_{i,l}^{[n]}\Lambda_nV(s)\mathcal{D}^{[n]}(\mathrm{d}s)\right| \tag{5.91}$$

$$\leq dn^d\frac{L}{n}\int_{\Sigma^{[n]}}(s_i+s_j)\mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.92}$$

Here, the factor dn^d comes from the number of edges in the grid. Since $\int_{\Sigma^{[n]}}(s_i+s_j)\mathcal{D}^{[n]}(\mathrm{d}s)=2/n^d$, we have

$$\left|\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\int_{\Sigma^{[n]}}\int_{\Sigma^{[n]}}\left(\frac{2\alpha+1}{\alpha}\cdot\frac{s_is_j}{s_i+s_j}-\frac{s_i+s_j}{2}\right)\partial_{i,l}^{[n]}\Lambda_nU(s)\partial_{i,l}^{[n]}\Lambda_nV(s)\mathcal{D}^{[n]}(\mathrm{d}s)\right| \tag{5.93}$$

$$\leq\frac{2Ld}{n}\to 0\quad\text{as } n\to\infty. \tag{5.94}$$

Therefore, the energy with the harmonic mean choice converges to the same limit as the energy with the arithmetic mean choice.

Step 2.2. Convergence of the energy with the arithmetic mean choice. We now show that the energy with the arithmetic mean θ converges to the desired limit. We first write down the energy explicitly. For notational ease, we write $\psi_{i,l}(s)=\partial_{i,l}^{[n]}\Lambda_nU(s)\partial_{i,l}^{[n]}\Lambda_nV(s)$. We then have the energy with the arithmetic mean choice as

$$\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\int_{\Sigma^{[n]}}\frac{s_i+s_{i+e_l}}{2}\psi_{i,l}(s)\mathcal{D}^{[n]}(\mathrm{d}s)=\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\int_{\Sigma^{[n]}}s_i\frac{\psi_{i,l}(s)+\psi_{i-e_l,l}(s)}{2}\mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.95}$$

$$=\int_{\mathbb{P}(\mathbb{T}^d)}\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\frac{\psi_{i,l}(\pi^{[n]}(\mu))+\psi_{i-e_l,l}(\pi^{[n]}(\mu))}{2}\mu(C_i^{[n]})\mathcal{D}(\mathrm{d}\mu) \tag{5.96}$$

$$=\int_{\mathbb{P}(\mathbb{T}^d)}\int_{\mathbb{T}^d}B_n(\mu,x)\mu(dx)\mathcal{D}(\mathrm{d}\mu) \tag{5.97}$$

where

$$B_n(\mu,x)=\sum_{l=1}^{d}\frac{\psi_{i,l}(\pi^{[n]}(\mu))+\psi_{i-e_l,l}(\pi^{[n]}(\mu))}{2},\quad x\in C_i^{[n]}. \tag{5.98}$$

In the above equality chain, for the first equality, we have used the adjoint operator of the arithmetic mean operator $s\mapsto\left(\frac{s_i+s_{i+e_l}}{2}\right)_i$, which is $s\mapsto\left(\frac{s_i+s_{i-e_l}}{2}\right)_i$. Our aim now is to show that $B_n(\mu,x)$ converges to $\langle DU(\mu)(x),DV(\mu)(x)\rangle$ uniformly in μ and x as $n\to\infty$. We first recall that

$$\partial_{i,l}^{[n]}\Lambda_nU(s)=\left(\nabla F(A_f(s))\cdot\frac{f_{*,j}-f_{*,i}}{h}\right) \tag{5.99}$$

$$=\sum_{k=1}^{m}\partial_kF(A_f(s))\frac{f_k(x_{i+e_l}^{[n]})-f_k(x_i^{[n]})}{h}. \tag{5.100}$$

Since f_k is smooth, by Taylor expansion, we have $\frac{f_k(x_{i+e_l}^{[n]})-f_k(x_i^{[n]})}{h}\to\nabla f_k(x)\cdot e_l$ uniformly in x as $n\to\infty$. Moreover, since we have seen earlier that $W_2(\iota^{[n]}\circ\pi_n(\mu),\mu)\to 0$ uniformly in μ as

$n \to \infty$, we have $A_f(\pi^{[n]}(\mu)) \to (\mu(f_1), \dots, \mu(f_m))$ uniformly in μ as $n \to \infty$. Overall, we have that $\psi_{i,l}(\pi^{[n]}(\mu))$ converges to

$$\left(\sum_{k=1}^{m} \partial_k F(\mu(f_1), \dots, \mu(f_m)) \nabla f_k(x) \cdot e_l\right) \left(\sum_{k=1}^{m} \partial_k G(\mu(g_1), \dots, \mu(g_m)) \nabla g_k(x) \cdot e_l\right) \tag{5.101}$$

uniformly in μ and x as $n \to \infty$. Summing over l, we have that $B_n(\mu, x)$ converges to $\langle DU(\mu)(x), DV(\mu)(x)\rangle$ uniformly in μ and x as $n \to \infty$. Therefore, by dominated convergence theorem, we have

$$\lim_{n\to\infty} \int_{\mathbb{P}(\mathbb{T}^d)} \int_{\mathbb{T}^d} B_n(\mu, x)\mu(dx)\mathcal{D}(\mathrm{d}\mu) = \int_{\mathbb{P}(\mathbb{T}^d)} \int_{\mathbb{T}^d} \langle DU(\mu)(x), DV(\mu)(x)\rangle \mu(dx)\mathcal{D}(\mathrm{d}\mu) = \mathcal{E}(U, V). \tag{5.102}$$

□

5.4. **Mosco liminf via Palm–trace duality.** We now prove the weak liminf inequality for the Eulerian forms. The proof is organized so that the main text contains only the structural ingredients. The detailed trace and Palm arguments are deferred to Section A.

The obstruction for the arithmetic mobility is the following. Conditional on the total two-cell mass ($r = s_i + s_j$), the split variable ($x = s_i/(s_i + s_j)$) has the small-parameter law Beta(α_n, α_n), and hence concentrates near the endpoints ($x = 0$ and $x = 1$). The arithmetic mean θ is completely independent to the movement of x given r, so a high slope in the middle x region is not penalized by the energy, and hence the energy of the limit function is underestimated. This is exactly the mechanism that is demonstrated in the counterexample in Proposition 5.2: χ has a slope smaller than 1, i.e., linear case, at the endpoints, but to compensate for the small mobility in the middle, χ has a slope larger than 1 in the middle region, which won't be penalized and leads to an energy smaller than the desired limit.

The renormalized harmonic mobility corrects precisely this endpoint capacity. In fact, using the same variables r, x, the harmonic θ is equal to $rx(1-x)$ up to a constant, which attains the maximum at $x = 1/2$. Therefore, the harmonic mean penalizes the high slope in the middle region, correcting the underestimation of the energy. The proof below makes this correction visible by replacing the edge gradient $\partial_{i,l}^{[n]}$ by an endpoint trace gradient $\nabla_{n,e}^{\mathrm{tr}}$. Throughout this subsection we write

$$h = \frac{1}{n}, \qquad \alpha_n = \frac{\beta}{n^d} = \beta h^d, \tag{5.103}$$

and we use the renormalized harmonic mobility

$$\theta_{i,l}^{[n]}(s) = \frac{2\alpha_n + 1}{\alpha_n} \frac{s_i s_{i+e_l}}{s_i + s_{i+e_l}}, \qquad \theta_{i,l}^{[n]}(s) = 0 \text{ if } s_i = s_{i+e_l} = 0. \tag{5.104}$$

Let

$$J_n : H^{[n]} \longrightarrow H = L^2(P(\mathbb{T}^d), \mathcal{D}), \qquad J_n F = F \circ \pi^{[n]}. \tag{5.105}$$

Thus convergence in the generalized Hilbert sense can be read after applying the isometric embedding J_n.

Endpoint trace variables. Fix an oriented edge $e = (i, l)$ and put $j = i + e_l$. On the two endpoint masses write

$$r = s_i + s_j, \qquad x = \frac{s_i}{s_i + s_j} \in [0, 1]. \tag{5.106}$$

Under $\mathcal{D}^{[n]}$, conditional on the variable $y = \big(r, (s_k)_{k\notin\{i,j\}}\big)$, x has law Beta(α_n, α_n). Let $\lambda_{n,e}$ be the law of y and define the edge Palm probability

$$\mathsf{P}_{n,e}(dy) = \frac{r}{2h^d}\, \lambda_{n,e}(dy). \tag{5.107}$$

For $a \in [0,1]$, let $s^{e,a}(y)$ denote the mass vector obtained from y by putting $s_i = ra$ and $s_j = r(1-a)$. Now consider

$$T^+_{n,e}F(y) = 2\int_0^1 aF(s^{e,a}(y))q_{\alpha_n}(a)\,da, \qquad T^-_{n,e}F(y) = 2\int_0^1 (1-a)F(s^{e,a}(y))q_{\alpha_n}(a)\,da, \tag{5.108}$$

where q_α is the Beta(α,α) density. These values are averages of F around each endpoint: $T^+_{n,e}$ morally evaluates the value of F when masses are concentrated at i, and $T^-_{n,e}$ when masses are concentrated at j. Then, the endpoint trace gradient is defined as

$$\nabla^{\mathrm{tr}}_{n,e}F(y) = \frac{T^+_{n,e}F(y) - T^-_{n,e}F(y)}{rh}. \tag{5.109}$$

This quantity measure how much F changes when mass was moved from the j-cell to the i-cell, which is the movement mostly invisible to the arithmetic mean θ. As the following lemma will show, the harmonic mean θ controls this trace gradient, and the trace gradient has a good discrete adjoint that converges to the Palm divergence in the limit. This is the key to the liminf inequality.

For edge-indexed functions $A = (A_e)$ and $C = (C_e)$ define

$$\langle A, C\rangle_{\mathrm{tr},n} = \sum_e h^d \int A_e(y)C_e(y)\,\mathsf{P}_{n,e}(dy), \qquad \|A\|^2_{\mathrm{tr},n} = \langle A, A\rangle_{\mathrm{tr},n}. \tag{5.110}$$

Lemma 5.6 (Endpoint capacity lower bound)**.** *There exists a sequence $\varepsilon_n \downarrow 0$ such that, for every $F \in \mathcal{D}(\mathcal{E}^{[n]})$,*

$$\mathcal{E}^{[n]}(F,F) \geq (1-\varepsilon_n)\,\|\nabla^{\mathrm{tr}}_n F\|^2_{\mathrm{tr},n}. \tag{5.111}$$

The first lemma says that the corrected harmonic mobility has the right control over the trace gradient. This is the step that fails for the arithmetic mobility: there the endpoint trace can be changed too cheaply. Once the trace gradient is controlled, we need test fields with a good discrete adjoint.

Let $\mathcal{A}$ denote the smooth cylinder algebra on $P(\mathbb{T}^d)$, and let $p(\mu,x) = \mu(\{x\})$ be the atom mass at x. Let $\mathscr{W}$ be the class of smooth cylinder vector fields of the form

$$W(\mu,x) = \sum_{a=1}^{A}\Gamma_a(\mu)\nabla\varphi_a(x), \qquad \Gamma_a \in \mathcal{A}, \quad \varphi_a \in C^\infty(\mathbb{T}^d). \tag{5.112}$$

For $W \in \mathscr{W}$ define its divergence by

$$B_W(\mu) = \int_{\mathbb{T}^d}\sum_{a=1}^{A}\left[\Gamma_a(\mu)\Delta\varphi_a(x) + p(\mu,x)D\Gamma_a(\mu,x)\cdot\nabla\varphi_a(x)\right]\mu(dx). \tag{5.113}$$

For an edge $e = (i,l)$, define the discrete trace test

$$(rW)_{n,e}(y) = -r\,W_l\big(\iota^{[n]}(s^{e,1/2}(y)), x^{[n],\mathrm{mid}}_{i,l}\big), \tag{5.114}$$

where $x^{[n],\mathrm{mid}}_{i,l}$ is the midpoint of the two neighboring cell centers $x^{[n]}_i$ and $x^{[n]}_{i+e_l}$. The minus sign compensates for the orientation convention in (5.109).

Lemma 5.7 (Discrete Palm adjoint and trace-test convergence)**.** *For every $W \in \mathscr{W}$, there are $B^W_n, R^W_n \in H^{[n]}$ such that, for every $F \in \mathcal{D}(\mathcal{E}^{[n]})$,*

$$\langle\nabla^{\mathrm{tr}}_n F, (rW)_n\rangle_{\mathrm{tr},n} = -\int_{\Sigma^{[n]}} F(s)B^W_n(s)\,\mathcal{D}^{[n]}(ds) + \int_{\Sigma^{[n]}} F(s)R^W_n(s)\,\mathcal{D}^{[n]}(ds). \tag{5.115}$$

Moreover,

$$J_nB^W_n \longrightarrow B_W \quad \textit{strongly in } H, \qquad \|R^W_n\|_{H^{[n]}} \longrightarrow 0, \tag{5.116}$$

and

$$\|(rW)_n\|_{\mathrm{tr},n}^2 \longrightarrow \int_{P(\mathbb{T}^d)} \int_{\mathbb{T}^d} |p(\mu,x)W(\mu,x)|^2\,\mu(dx)\mathcal{D}(d\mu). \tag{5.117}$$

This lemma is the discrete-to-continuum bridge. Its point is not only that a discrete integration by parts is possible, but that the adjoint B_n^W converges strongly to the bounded Palm divergence B_W. Hence weak convergence of F_n is enough to pass to the limit when testing against B_n^W.

Lemma 5.8 (Continuum Palm duality)**.** *Assume $d \geq 2$. For every $U \in H$, let*

$$\overline{\mathcal{E}}(U,U) = \begin{cases} \mathcal{E}(U,U), & U \in \mathcal{D}(\mathcal{E}), \\ +\infty, & U \notin \mathcal{D}(\mathcal{E}). \end{cases}$$

Then

$$\overline{\mathcal{E}}(U,U) = \sup_{W\in\mathscr{W}} \left\{ -2\int_{P(\mathbb{T}^d)} U(\mu)B_W(\mu)\,\mathcal{D}(d\mu) - \int_{P(\mathbb{T}^d)}\int_{\mathbb{T}^d} |p(\mu,x)W(\mu,x)|^2\,\mu(dx)\mathcal{D}(d\mu) \right\}. \tag{5.118}$$

The last lemma says that the mass-weighted tests are not a smaller energy: after taking the supremum over $W \in \mathscr{W}$, they recover the full closed Wasserstein gradient.

Proposition 5.9 (Weak liminf inequality)**.** *Assume $d \geq 2$. If $F_n \in H^{[n]}$ converges weakly to $U \in H$ in the generalized Hilbert sense, then*

$$\overline{\mathcal{E}}(U,U) \leq \liminf_{n\to\infty} \mathcal{E}^{[n]}(F_n,F_n). \tag{5.119}$$

Proof. If the right-hand side is $+\infty$, there is nothing to prove. Passing to a subsequence that realizes the liminf, we may therefore assume $F_n \in \mathcal{D}(\mathcal{E}^{[n]})$ and $\sup_n \mathcal{E}^{[n]}(F_n,F_n) < \infty$.

By Theorem 5.6, choose $c_n \to 1$ such that

$$\mathcal{E}^{[n]}(F_n,F_n) \geq c_n \|\nabla_n^{\mathrm{tr}} F_n\|_{\mathrm{tr},n}^2. \tag{5.120}$$

Fix $W \in \mathscr{W}$. Completing the square in the trace Hilbert space gives

$$c_n \|\nabla_n^{\mathrm{tr}} F_n\|_{\mathrm{tr},n}^2 \geq 2\langle \nabla_n^{\mathrm{tr}} F_n, (rW)_n\rangle_{\mathrm{tr},n} - c_n^{-1}\|(rW)_n\|_{\mathrm{tr},n}^2. \tag{5.121}$$

By Theorem 5.7,

$$\langle \nabla_n^{\mathrm{tr}} F_n, (rW)_n\rangle_{\mathrm{tr},n} = -\int_{\Sigma^{[n]}} F_n B_n^W \,d\mathcal{D}^{[n]} + \int_{\Sigma^{[n]}} F_n R_n^W \,d\mathcal{D}^{[n]}.$$

The first term converges to $-\int U B_W \,d\mathcal{D}$, because F_n converges weakly and B_n^W converges strongly. The second term vanishes: the energy bound and weak convergence give a uniform $H^{[n]}$-bound on F_n, while $R_n^W \to 0$ in $H^{[n]}$. Hence

$$\lim_{n\to\infty} \langle \nabla_n^{\mathrm{tr}} F_n, (rW)_n\rangle_{\mathrm{tr},n} = -\int_{P(\mathbb{T}^d)} U(\mu)B_W(\mu)\,\mathcal{D}(d\mu). \tag{5.122}$$

The norm convergence in Theorem 5.7 also gives

$$\lim_{n\to\infty} c_n^{-1}\|(rW)_n\|_{\mathrm{tr},n}^2 = \int_{P(\mathbb{T}^d)}\int_{\mathbb{T}^d} |p(\mu,x)W(\mu,x)|^2\,\mu(dx)\mathcal{D}(d\mu). \tag{5.123}$$

Combining (5.120)–(5.123) yields, for every $W \in \mathscr{W}$,

$$\begin{aligned} \liminf_{n\to\infty} \mathcal{E}^{[n]}(F_n,F_n) \geq -2\int_{P(\mathbb{T}^d)} U(\mu)B_W(\mu)\,\mathcal{D}(d\mu) \\ - \int_{P(\mathbb{T}^d)}\int_{\mathbb{T}^d} |p(\mu,x)W(\mu,x)|^2\,\mu(dx)\mathcal{D}(d\mu). \end{aligned}$$

Taking the supremum over $W \in \mathscr{W}$ and using Theorem 5.8 gives (5.119). $\square$

We will now show that $\mathcal{E}^{[n]}$ is in fact a closable Dirichlet form on $H^{[n]}$.

Theorem 5.10. *Let $\mathcal{C}^{[n]}$ be the set of smooth functions on $\Sigma^{[n]}$. Then the symmetric bilinear form $(\mathcal{E}^{[n]}, \mathcal{C}^{[n]})$ is closable, and the closure denoted by $(\mathcal{E}^{[n]}, \mathcal{F}^{[n]})$ is a regular conservative strongly local Dirichlet form on $H^{[n]}$. regular: has a core, and implies Hunt process, Chapter 6 6.2.1 Conservativeness: includes 1 in the domain, and $\mathcal{E}^{[n]}(1,1) = 0$. Implies that it has no killing i.e. does not got to cemetry. S1.4, p27, S4.1, 89-90 strongly local: $\mathcal{E}^{[n]}(f,g) = 0$ if f is constant on the support of g. Implies that the associated process is a diffusion, i.e., has continuous paths. C4 114, theorem 4.5.1, Chapter 6 p184-185, theorem 6.2.2*

Proof. □

By the above theorem we obtain a unique diffusion process associated with the Dirichlet form $\mathcal{E}^{[n]}$.

Corollary 5.11. *something like "We have a unique diffusion with no killing associated with the Dirichlet form $\mathcal{E}^{[n]}$."*

Proof. Cite fukushima's book for each property of the Dirichlet form and the corresponding property of the process. □

5.5. **The derivation of the generator and the associated SDE.** We now derive the generator associated with the Dirichlet form $\mathcal{E}^{[n]}$, i.e., the non-positive self-adjoint operator $L^{[n]}$ that corresponds one-to-one to the Dirichlet form $\mathcal{E}^{[n]}$ [4, p. 19] by the relation

$$\mathcal{D}(L^{[n]}) \subset \mathcal{D}(\mathcal{E}^{[n]}), \qquad \mathcal{E}^{[n]}(F,G) = -\langle L^{[n]}F, G\rangle_{H^{[n]}}, \quad \forall F \in \mathcal{D}(L^{[n]}), G \in \mathcal{D}(\mathcal{E}^{[n]}). \tag{5.124}$$

Here, $\mathcal{D}(\cdot)$ the domain of the operator or the Dirichlet form. The generator will allow us to derive the SDE satisfied by the diffusion process by Fukushima's decomposition.

5.5.1. *The generator formula for smooth functions.* We first derive the following formula for the smooth functions. Let $\rho^{[n]}$ be the density of $\mathcal{D}^{[n]}$ with respect to the Lebesgue measure on $\Sigma^{[n]}$. We have the following formula for $L^{[n]}F$ for any smooth function F:

Proposition 5.12. *For any smooth functions F on $\Sigma^{[n]}$, we have that $F \in \mathcal{D}(L^{[n]})$ and*

$$L^{[n]}F(s) = \frac{1}{\rho^{[n]}(s)} \sum_{i\in\mathcal{I}^{[n]}} \sum_{l=1}^{d} \partial_{i,l}^{[n]} \left(\theta_{i,l}(s)\rho^{[n]}(s)\partial_{i,l}^{[n]}F(s)\right) \tag{5.125}$$

$$= \sum_{i\in\mathcal{I}^{[n]}} \sum_{l=1}^{d} \left(\theta_{i,l}(s)\partial_{i,l}^{[n]}\partial_{i,l}^{[n]} + \frac{2\alpha+1}{h}\frac{s_i - s_j}{s_i + s_j}\partial_{i,l}^{[n]}\right) F(s) \tag{5.126}$$

Proof. Recall that the Dirichlet form for smooth functions F, G is given by

$$\mathcal{E}^{[n]}(F,G) = \sum_{i\in\mathcal{I}^{[n]}} \sum_{l=1}^{d} \int_{\Sigma^{[n]}} \theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\partial_{i,l}^{[n]}G(s)\mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.127}$$

We fix an edge (i,l) and write $j = i + e_l$. Define

$$v_{i,l} = \frac{e_j - e_i}{h} \tag{5.128}$$

so that $\partial_{i,l}^{[n]}F(s) = \nabla F(s) \cdot v_{i,l}$ where ∇ represents the tangential gradient on $\Sigma^{[n]}$. We consider the term involving the edge (i,l) in the energy. To avoid singularity with zero mass, we consider a subset of $\Sigma^{[n]}$ given by $\Sigma_\epsilon^{[n]} = \{s \in \Sigma^{[n]} : \forall i, s_i \geq \epsilon\}$ for $\epsilon > 0$ and consider the integral

$$\int_{\Sigma_\epsilon^{[n]}} \theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\partial_{i,l}^{[n]}G(s)\mathcal{D}^{[n]}(\mathrm{d}s) = \int_{\Sigma_\epsilon^{[n]}} \theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\partial_{i,l}^{[n]}G(s)\rho^{[n]}(s)\mathrm{d}s. \tag{5.129}$$

Step 1. Integration by parts. We first apply the product rule to obtain

$$\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\partial_{i,l}^{[n]}G(s)\rho^{[n]}(s) = \left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\partial_{i,l}^{[n]}G(s) \tag{5.130}$$

$$= \partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)\right) - G(s)\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right) \tag{5.131}$$

$$= \nabla\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)\right)\cdot v_{i,l} - G(s)\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right) \tag{5.132}$$

$$= \operatorname{div}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\right) - G(s)\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right). \tag{5.133}$$

Here, $\div$ is the tangential divergence on $\Sigma^{[n]}$. We integrate both sides over $\Sigma_\epsilon^{[n]}$ and use the divergence theorem for Lipschitz domain [9, Theorem 3.34] to obtain

$$\int_{\partial\Sigma_\epsilon^{[n]}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\cdot n(s)\mathrm{d}\sigma(s) - \int_{\Sigma_\epsilon^{[n]}}G(s)\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathrm{d}s \tag{5.134}$$

Here, $n(s)$ is the outward normal vector at s, σ is the surface measure on $\partial\Sigma_\epsilon^{[n]}$, and $\partial\sigma_\epsilon^{[n]}$ is a boundary relative to $\sigma^{[n]}$. We will now prove that the boundary term would go to zero $\epsilon \to 0$.

Step 2. Boundary term vanishes. We decompose the boundary into the following faces:

$$\Gamma_{r,\epsilon} = \{s \in \Sigma_\epsilon^{[n]} : s_r = \epsilon\}, \quad \partial\Sigma_\epsilon^{[n]} = \bigcup_{r\in\mathcal{I}^{[n]}}\Gamma_{r,\epsilon}. \tag{5.135}$$

Note that the intersection of any two faces has zero surface measure since it has codimension at least 2 due to the two active constraints. Therefore the boundary integral is the sum of integrals on $\Gamma_{r,\epsilon}$ for $r \in \mathcal{I}^{[n]}$:

$$\int_{\partial\Sigma_\epsilon^{[n]}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\cdot n(s)\mathrm{d}\sigma(s) = \sum_{r\in\mathcal{I}^{[n]}}\int_{\Gamma_{r,\epsilon}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\cdot n(s)\mathrm{d}\sigma(s). \tag{5.136}$$

On $\Gamma_{r,\epsilon}$, the outward normal vector in the ambient space $\mathbb{R}^{\mathcal{I}^{[n]}}$ is given by $-e_r$. The tangential outward normal vector $n(s)$ is the projection of $-e_r$ to the tangent space of $\Sigma^{[n]}$ at s. Since the tangent space of $\Sigma^{[n]}$ at s is given by $\{v \in \mathbb{R}^{\mathcal{I}^{[n]}} : \sum_{i\in\mathcal{I}^{[n]}} v_i = 0\}$, or the orthogonal complement of the vector $\mathbf{1} = (1, 1, \dots, 1)$, we have

$$n(s) = \frac{-e_r + \frac{1}{n^d}\mathbf{1}}{\|-e_r + \frac{1}{n^d}\mathbf{1}\|} = \frac{-e_r + \frac{1}{n^d}\mathbf{1}}{\sqrt{1 - \frac{1}{n^d}}}. \tag{5.137}$$

This implies that we have

$$v_{i,l}\cdot n(s) = \frac{1}{h}\frac{\delta_{i,r} - \delta_{j,r}}{\sqrt{1 - \frac{1}{n^d}}}. \tag{5.138}$$

Here, $\delta_{i,r}$ is the Kronecker delta. In particular, we have $v_{i,l}\cdot n(s) = 0$ for $r \notin \{i, j\}$, so the boundary integral only has contributions from $\Gamma_{i,\epsilon}$ and $\Gamma_{j,\epsilon}$. We now assume that $r \in \{i, j\}$ and consider $\theta_{i,l}(s)$. Recall that $\theta_{i,l}(s) = \frac{2\alpha+1}{\alpha}\cdot\frac{s_is_j}{s_i+s_j}$, where $\alpha = \frac{\beta}{n^d}$ is the parameter of the Dirichlet distribution $\mathcal{D}^{[n]}$. Since $s_i, s_j \geq 0$, we have

$$\frac{s_is_j}{s_i + s_j} \leq \min(s_i, s_j) \leq \epsilon \tag{5.139}$$

because at least one of s_i and s_j must be equal to ϵ on the boundary. Combined with

$$\rho^{[n]}(s) = \frac{\Gamma(\alpha n^d)}{\Gamma(\alpha)^{n^d}}\prod_{i\in\mathcal{I}^{[n]}} s_i^{\alpha-1} \tag{5.140}$$

we have that, since $s_r=\epsilon$ on $\Gamma_{r,\epsilon}$,

$$\theta_{i,l}(s)\rho^{[n]}(s)\le C\epsilon^{\alpha}\prod_{i\in\mathcal{I}^{[n]}\setminus\{r\}}s_i^{\alpha-1}. \tag{5.141}$$

where C is a constant independent of ϵ. Now use the fact that F, G are smooth so that themselves and derivatives are bounded, we have

$$\left|\int_{\Gamma_{r,\epsilon}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\cdot n(s)\mathrm{d}\sigma(s)\right|\le C\epsilon^{\alpha}\int_{\Gamma_{r,\epsilon}}\prod_{i\in\mathcal{I}^{[n]}\setminus\{r\}}s_i^{\alpha-1}\mathrm{d}\sigma(s). \tag{5.142}$$

where C has absorbed the bound of $\partial_{i,l}^{[n]}F(s)$, $G(s)$ and $v_{i,l}\cdot n(s)$. It remains to evaluate the integral on the right hand side. The set $\Gamma_{r,\epsilon}$ is contained in the set

$$\left\{s\in\Sigma_{\epsilon}^{[n]}\middle|s_r=\epsilon,\forall q,s_q\ge\epsilon,\sum_{i\in\mathcal{I}^{[n]}\setminus\{r\}}s_i=1-\epsilon\right\}. \tag{5.143}$$

Therefore, with change of variable

$$t_q=\frac{s_q}{1-\epsilon},\quad q\in\mathcal{I}^{[n]}\setminus\{r\},\sum_{q\in\mathcal{I}^{[n]}\setminus\{r\}}t_q=1, \tag{5.144}$$

we have

$$\int_{\Gamma_{r,\epsilon}}\prod_{i\in\mathcal{I}^{[n]}\setminus\{r\}}s_i^{\alpha-1}\mathrm{d}\sigma(s)=(1-\epsilon)^{(n^d-1)(\alpha-1)}\int_{\Delta_{1-\epsilon}^{[n]}}\prod_{i\in\mathcal{I}^{[n]}\setminus\{r\}}t_i^{\alpha-1}\mathrm{d}\sigma(t), \tag{5.145}$$

where $\Delta_{1-\epsilon}^{[n]}=\{t\in\mathbb{R}^{\mathcal{I}^{[n]}\setminus\{r\}}:t_i\ge 0,\sum_{i\in\mathcal{I}^{[n]}\setminus\{r\}}t_i=1\}$ is a simplex. The integral on the right hand side is finite since it is exactly the unnormalized density of a Dirichlet distribution with parameter α on the simplex. Therefore, we have

$$\left|\int_{\Gamma_{r,\epsilon}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\cdot n(s)\mathrm{d}\sigma(s)\right|\le C'\epsilon^{\alpha} \tag{5.146}$$

for some constant C' independent of ϵ. Therefore, we have

$$\left|\int_{\partial\Sigma_{\epsilon}^{[n]}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)G(s)v_{i,l}\cdot n(s)\mathrm{d}\sigma(s)\right|\le C''\epsilon^{\alpha}\to 0\quad\text{ as }\epsilon\to 0. \tag{5.147}$$

Step 3. Passing $\epsilon\to 0$. We have shown that the boundary term vanishes as $\epsilon\to 0$. We now show that we can apply the dominated convergence theorem to pass the limit $\epsilon\to 0$ in the other term:

$$-\int_{\Sigma_{\epsilon}^{[n]}}G(s)\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathrm{d}s=-\int_{\Sigma_{\epsilon}^{[n]}}G(s)\frac{1}{\rho^{[n]}(s)}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.148}$$

We note that $\rho^{[n]}(s)$ can be unbounded, so we will show that the whole integrand is bounded through algebraic manipulation. We first apply the product rule to obtain

$$\frac{1}{\rho^{[n]}(s)}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)=\theta_{i,l}(s)\partial_{i,l}^{[n]}\partial_{i,l}^{[n]}F(s)+\left(\partial_{i,l}^{[n]}\theta_{i,l}(s)+\theta_{i,l}(s)\partial_{i,l}^{[n]}\log\rho^{[n]}(s)\right)\partial_{i,l}^{[n]}F(s) \tag{5.149}$$

(5.150)

We first observe that

$$\partial_{i,l}\log\rho^{[n]}(s)=\frac{\alpha-1}{h}\left(\frac{1}{s_j}-\frac{1}{s_i}\right)=\frac{\alpha-1}{h}\frac{s_i-s_j}{s_is_j},\quad\partial_{i,l}^{[n]}\theta_{i,l}(s)=\frac{2\alpha+1}{\alpha h}\cdot\frac{s_i-s_j}{s_i+s_j}. \tag{5.151}$$

Therefore, we have

$$\partial_{i,l}^{[n]}\theta_{i,l}(s)+\theta_{i,l}(s)\partial_{i,l}^{[n]}\log\rho^{[n]}(s)=\frac{2\alpha+1}{\alpha h}\cdot\frac{s_i-s_j}{s_i+s_j}+\frac{2\alpha+1}{\alpha}\cdot\frac{s_is_j}{s_i+s_j}\cdot\frac{\alpha-1}{h}\frac{s_i-s_j}{s_is_j} \tag{5.152}$$

$$=\frac{2\alpha+1}{\alpha h}\cdot\frac{s_i-s_j}{s_i+s_j}\left(1+\alpha-1\right) \tag{5.153}$$

$$=\frac{2\alpha+1}{\alpha h}\cdot\frac{s_i-s_j}{s_i+s_j}\cdot\alpha=\frac{2\alpha+1}{h}\cdot\frac{s_i-s_j}{s_i+s_j}. \tag{5.154}$$

We note that $\left|\frac{s_i-s_j}{s_i+s_j}\right|\leq 1$ by the triangle inequality and the nonnegativity of s_i and s_j. Therefore, the term $\partial_{i,l}^{[n]}\theta_{i,l}(s)+\theta_{i,l}(s)\partial_{i,l}^{[n]}\log\rho^{[n]}(s)$ is bounded. Now, $G(s)$ and $\partial_{i,l}^{[n]}F(s)$ are also bounded since F and G are smooth. Finally, we note that $\theta_{i,l}(s)=\frac{2\alpha+1}{\alpha}\cdot\frac{s_is_j}{s_i+s_j}\leq\frac{2\alpha+1}{\alpha}\cdot\frac{s_i+s_j}{4}$ by AM-GM inequality, so $\theta_{i,l}(s)$ is also bounded. Therefore, the whole integrand is bounded, and we can apply the dominated convergence theorem to pass the limit $\epsilon\to 0$ to obtain

$$\lim_{\epsilon\to 0}-\int_{\Sigma_\epsilon^{[n]}}G(s)\frac{1}{\rho^{[n]}(s)}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.155}$$

$$=-\int_{\Sigma^{[n]}}G(s)\frac{1}{\rho^{[n]}(s)}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.156}$$

Step 4. Conclusion. We have shown that for any edge (i,l), we have

$$\int_{\Sigma^{[n]}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\partial_{i,l}^{[n]}G(s)\mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.157}$$

$$=-\int_{\Sigma^{[n]}}G(s)\frac{1}{\rho^{[n]}(s)}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.158}$$

Summing over all edges, we have

$$\mathcal{E}^{[n]}(F,G)=\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\int_{\Sigma^{[n]}}\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\partial_{i,l}^{[n]}G(s)\mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.159}$$

$$=-\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\int_{\Sigma^{[n]}}G(s)\frac{1}{\rho^{[n]}(s)}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathcal{D}^{[n]}(\mathrm{d}s) \tag{5.160}$$

$$=-\int_{\Sigma^{[n]}}G(s)\frac{1}{\rho^{[n]}(s)}\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right)\mathcal{D}^{[n]}(\mathrm{d}s). \tag{5.161}$$

Denote

$$\mathscr{L}^{[n]}F(s)=\frac{1}{\rho^{[n]}(s)}\sum_{i\in\mathcal{I}^{[n]}}\sum_{l=1}^{d}\partial_{i,l}^{[n]}\left(\theta_{i,l}(s)\partial_{i,l}^{[n]}F(s)\rho^{[n]}(s)\right). \tag{5.162}$$

so that

$$\mathcal{E}^{[n]}(F,G)=-\int_{\Sigma^{[n]}}G(s)\mathscr{L}^{[n]}F(s)\mathcal{D}^{[n]}(\mathrm{d}s)=-\langle\mathscr{L}^{[n]}F,G\rangle_{H^{[n]}}. \tag{5.163}$$

We have this equality for any smooth F,G. Since the space of smooth functions forms a core, i.e., it is dense in $\mathcal{D}(\mathcal{E}^{[n]})$ under the norm $\|F\|_{\mathcal{E}_1^{[n]}}=\sqrt{\|F\|_{H^{[n]}}^2+\mathcal{E}^{[n]}(F,F)}$ [4, p. 5], we have the same equality for smooth F and $G\in\mathcal{D}(\mathcal{E}^{[n]})$. Indeed, for a sequence G_k of smooth functions converging to G in $\|\cdot\|_{\mathcal{E}_1^{[n]}}$, we have $G_k \to G$ in $\mathcal{H}^{[n]}$ too, so by using the fact that $\mathscr{L}^{[n]}F$ is in $H^{[n]}$ since the integrand is bounded by the arguments in Step 3, we have

$$|\langle \mathscr{L}^{[n]}F, G\rangle_{H^{[n]}} - \langle \mathscr{L}^{[n]}F, G_k\rangle_{H^{[n]}}| \le \|\mathscr{L}^{[n]}F\|_{H^{[n]}}\|G-G_k\|_{H^{[n]}} \to 0 \tag{5.164}$$

as $k\to\infty$. Moreover, since $G_k \to G$ in $\|\cdot\|_{\mathcal{E}_1^{[n]}}$, we have $\mathcal{E}^{[n]}(G-G_k, G-G_k)\to 0$ as $k\to\infty$. Since $\mathcal{E}^{[n]}(F,F)$ is finite by smoothness, we have

$$|\mathcal{E}^{[n]}(F,G) - \mathcal{E}^{[n]}(F,G_k)| \le \sqrt{\mathcal{E}^{[n]}(F,F)}\sqrt{\mathcal{E}^{[n]}(G-G_k, G-G_k)} \to 0 \tag{5.165}$$

Therefore we have $\mathcal{E}^{[n]}(F,G) = -\langle \mathscr{L}^{[n]}F, G\rangle_{H^{[n]}}$ for any smooth F and $G\in\mathcal{D}(\mathcal{E}^{[n]})$. From here, we now show that $F\in\mathcal{D}(L^{[n]})$. Indeed, consider the resolvent $R_\alpha^{[n]}$, which has the domain of all of $H^{[n]}$, $R^{[n]}(H^{[n]}) = \mathcal{D}(L^{[n]})$ and satisfies [4, p. 14-16]

$$\mathcal{E}_\alpha^{[n]}(R_\alpha^{[n]}F, G) = \langle F, G\rangle_{H^{[n]}}, \quad F\in\mathcal{H}^{[n]}, \forall G\in\mathcal{D}(\mathcal{E}^{[n]}). \tag{5.166}$$

Here, $\mathcal{E}_\alpha^{[n]}(F,G) = \mathcal{E}^{[n]}(F,G) + \alpha\langle F,G\rangle_{H^{[n]}}$. We have shown that $\mathcal{E}^{[n]}(F,G) = -\langle \mathscr{L}^{[n]}F, G\rangle_{H^{[n]}}$ for any smooth F and $G\in\mathcal{D}(\mathcal{E}^{[n]})$, so we have

$$\mathcal{E}_\alpha^{n}(F,G) = -\langle \mathscr{L}^{[n]}F, G\rangle_{H^{[n]}} + \alpha\langle F,G\rangle_{H^{[n]}} = \langle \alpha F - \mathscr{L}^{[n]}F, G\rangle_{H^{[n]}} = \mathcal{E}_\alpha^{[n]}(R_\alpha^{[n]}(\alpha F - \mathscr{L}^{[n]}F), G). \tag{5.167}$$

Therefore we have $F = R_\alpha^{[n]}(\alpha F - \mathscr{L}^{[n]}F)$, so $F\in R^{[n]}(H^{[n]}) = \mathcal{D}(L^{[n]})$. Now $\mathcal{E}^{[n]}(F,G) = -\langle \mathscr{L}^{[n]}F, G\rangle_{H^{[n]}} = -\langle L^{[n]}F, G\rangle_{H^{[n]}}$ for any $G\in\mathcal{D}(\mathcal{E}^{[n]})$, so we have $L^{[n]}F = \mathscr{L}^{[n]}F$.

□

5.5.2. *The closed form SDE and the martingale problem.* Using the generator formula for the smooth function, we can identify the SDE satisfied by the diffusion process associated with $\mathcal{E}^{[n]}$ through Fukushima's decomposition. We first recall Fukushima's decomposition.

Theorem 5.13 (Corollary of Theorem 5.2.2., [4])**.** *Let S be a locally compact separable Hausdorff space, and let m be a positive Radon measure on S with full support. Let $(\mathcal{E},\mathcal{F})$ be a regular conservative symmetric Dirichlet form on $L^2(S,m)$ and $\{X_t\}_{t\ge 0}$ be the Hunt process associated with $(\mathcal{E},\mathcal{F})$ and L be a generator associated with $(\mathcal{E},\mathcal{F})$. Then for any continuous $u\in\mathcal{D}(L)$, the process*

$$M_t^{[u]} = u(X_t) - u(X_0) - \int_0^t Lu(X_s)\,ds, \quad t\ge 0 \tag{5.168}$$

is martingale under $\mathbb{P}_x$ for quasi-every $x\in S$.

We will use this theorem to $p_i^{[n]} : \Sigma^{[n]} \to \mathbb{R}, p_i^{[n]}(s) = s_i$, which is the projection to the i-th coordinate, for $i\in\mathcal{I}^{[n]}$. The result is applicable to $p_i^{[n]}$ because it is a smooth function, so it is in the domain of the generator $L^{[n]}$ by Proposition 5.12. Therefore, we have the decomposition:

$$dS_i^{[n]}(t) = L^{[n]}p_i^{[n]}(S^{[n]}(t))dt + dM_t^{[p_i^{[n]}]}, \quad i\in\mathcal{I}^{[n]}, \tag{5.169}$$

where $S_i^{[n]}(t) = p_i^{[n]}(S^{[n]}(t))$ is the i-th coordinate of the diffusion process $S^{[n]}(t)$ defined by our Eulerian formulation, and $M_t^{[p_i^{[n]}]}$ is a martingale. We will calculate each term in the above SDE to obtain a closed-form SDE for $S^{[n]}(t)$. We first calculate the drift term $L^{[n]}p_i^{[n]}(s)$. We have the following result:

Proposition 5.14. *For any $i\in\mathcal{I}^{[n]}$ and $s\in\Sigma^{[n]}$, we have*

$$L^{[n]}p_i^{[n]}(s) = \sum_{l=1}^d b_{i-e_l,l}(s) - \sum_{l=1}^d b_{i,l}(s), \quad b_{i,l}(s) = \frac{2\alpha+1}{h^2}\cdot\frac{s_i - s_{i+e_l}}{s_i + s_{i+e_l}}. \tag{5.170}$$

with the convention that $b_{i,l}(s) = 0$ *if both* s_i *and* s_{i+e_l} *are zero.*

Proof. This follows from the generator formula in Proposition 5.12 and the convention that $\theta_{i,l}(s) = \frac{2\alpha+1}{\alpha} \cdot \frac{s_i s_{i+e_l}}{s_i + s_{i+e_l}}$ is zero if both s_i and s_{i+e_l} are zero. □

We will now consider the martingale term $M_t^{[p_i^{[n]}]}$. To this end, we calculate the quadratic covariation of $M_t^{[p_i^{[n]}]}$ and $M_t^{[p_j^{[n]}]}$ for $i, j \in \mathcal{I}^{[n]}$ and use the integral representation of martingale [7, Theorem 19.13]. By [11, Proposition VIII.3.3], we have

$$\langle M^{[p_i^{[n]}]}, M^{[p_j^{[n]}]} \rangle_t = \int_0^t \Gamma(p_i^{[n]}, p_j^{[n]})(S^{[n]}(s))ds, \quad t \geq 0, \tag{5.171}$$

where Γ is the carré du champ operator defined by $\Gamma(u, v) = (L(uv) - uLv - vLu)$ for $u, v \in \mathcal{D}(L)$ and $\langle , \rangle$ denotes the quadratic covariation. Therefore, we calculate the carré du champ operator $\Gamma(p_i^{[n]}, p_j^{[n]})$. We have the following result:

Proposition 5.15. *For any smooth functions* $f, g \in \mathcal{D}(L^{[n]})$*, we have*

$$\Gamma(f, g)(s) = \sum_{i \in \mathcal{I}^{[n]}} \sum_{l=1}^{d} 2\theta_{i,l}(s) \partial_{i,l}^{[n]} f(s) \partial_{i,l}^{[n]} g(s). \tag{5.172}$$

In particular, for any $p, q \in \mathcal{I}^{[n]}$ *and* $s \in \Sigma^{[n]}$*, we have*

$$\Gamma(p_p^{[n]}, p_q^{[n]})(s) = \sum_{i \in \mathcal{I}^{[n]}} \sum_{l=1}^{d} \frac{2\theta_{i,l}(s)}{h^2} \cdot (\delta_{p,i+e_l} - \delta_{p,i}) (\delta_{q,i+e_l} - \delta_{q,i}). \tag{5.173}$$

Proof. This follows from the generator formula in Proposition 5.12 and the definition of Γ. □

The above proposition provides the enough information to identify the martingale term $M_t^{[p_i^{[n]}]}$ as a stochastic integral with respect to a Brownian motion. Using [7, Theorem 19.13], we have that, after possibly enlarging the probability space, there exists a Brownian motion $W^{[n]}(t) = (W_{i,l}^{[n]}(t))_{i \in \mathcal{I}^{[n]}, l=1,\ldots,d}$ in $\mathbb{R}^{n^d}$ such that, for any $p \in \mathcal{I}^{[n]}$,

$$dM_t^{[p_p^{[n]}]} = \sum_{i \in \mathcal{I}^{[n]}} \sum_{l=1}^{d} \frac{\sqrt{2\theta_{i,l}(S^{[n]}(t))}}{h} \cdot (\delta_{p,i+e_l} - \delta_{p,i}) \, dW_{i,l}^{[n]}(t) \tag{5.174}$$

$$= \sum_{l=1}^{d} \frac{\sqrt{2\theta_{p-e_l,l}(S^{[n]}(t))}}{h} dW_{p-e_l,l}^{[n]}(t) - \sum_{l=1}^{d} \frac{\sqrt{2\theta_{p,l}(S^{[n]}(t))}}{h} dW_{p,l}^{[n]}(t). \tag{5.175}$$

Combining the above calculations for the drift and diffusion terms, we have the following SDE for $S_i^{[n]}(t)$:

$$\begin{aligned} dS_i^{[n]}(t) &= \sum_{l=1}^{d} b_{i-e_l,l}(S^{[n]}(t))dt - \sum_{l=1}^{d} b_{i,l}(S^{[n]}(t))dt \\ &\quad + \sum_{l=1}^{d} \frac{\sqrt{2\theta_{i-e_l,l}(S^{[n]}(t))}}{h} dW_{i-e_l,l}^{[n]}(t) - \sum_{l=1}^{d} \frac{\sqrt{2\theta_{i,l}(S^{[n]}(t))}}{h} dW_{i,l}^{[n]}(t). \end{aligned} \tag{5.176}$$

Appendix A. Details for the Palm–trace liminf proof

This appendix contains the proofs of the three structural lemmas used in Section 5.4. I keep the notation of that subsection.

A.1. **Endpoint capacity.** For $0<\alpha<1$ set

$$q_\alpha(a)=\frac{1}{B(\alpha,\alpha)}a^{\alpha-1}(1-a)^{\alpha-1},\qquad m_\alpha(a)=\frac{2\alpha+1}{\alpha}a(1-a),\qquad w_\alpha=m_\alpha q_\alpha. \tag{A.1}$$

Then

$$R_\alpha:=\int_0^1\frac{da}{w_\alpha(a)}=\frac{\alpha B(\alpha,\alpha)}{2\alpha+1}B(1-\alpha,1-\alpha)\longrightarrow 2. \tag{A.2}$$

Proof of Theorem 5.6. Fix an edge $e=(i,l)$ and put $j=i+e_l$. Let $\eta_{\alpha,+}(da)=2aq_\alpha(a)da$ and $\eta_{\alpha,-}(da)=2(1-a)q_\alpha(a)da$. For every smooth $\psi:[0,1]\to\mathbb{R}$,

$$\int_0^1 w_\alpha(a)|\psi'(a)|^2\,da\geq\frac{\left|\int\psi\,d\eta_{\alpha,+}-\int\psi\,d\eta_{\alpha,-}\right|^2}{R_\alpha}. \tag{A.3}$$

Indeed, if $f_\alpha=d\eta_{\alpha,+}/da-d\eta_{\alpha,-}/da$ and $S_\alpha(t)=\int_0^t f_\alpha(a)da$, then $S_\alpha(0)=S_\alpha(1)=0$ and $|S_\alpha|\leq 1$. Integration by parts and Cauchy–Schwarz with the weight w_α give (A.3).

Now disintegrate $\mathcal{D}^{[n]}$ along the edge split. In the coordinates $r=s_i+s_j$, $a=s_i/(s_i+s_j)$ and collapsed variable y, one has

$$\mathcal{D}^{[n]}(ds)=\lambda_{n,e}(dy)q_{\alpha_n}(a)da.$$

For $\psi_y(a)=F(s^{e,a}(y))$,

$$\psi_y'(a)=r(\partial_{s_i}F-\partial_{s_j}F)(s^{e,a}(y))=-rh\,\partial_{i,l}^{[n]}F(s^{e,a}(y))$$

and

$$\theta_{i,l}^{[n]}(s^{e,a}(y))=r\,m_{\alpha_n}(a).$$

Therefore (A.3) gives

$$\int_{\Sigma^{[n]}}\theta_{i,l}^{[n]}|\partial_{i,l}^{[n]}F|^2\,d\mathcal{D}^{[n]}\geq\frac{2h^d}{R_{\alpha_n}}\int|\nabla_{n,e}^{\mathrm{tr}}F(y)|^2\,\mathsf{P}_{n,e}(dy).$$

Summing over the oriented edges gives

$$\mathcal{E}^{[n]}(F,F)\geq\frac{2}{R_{\alpha_n}}\|\nabla_n^{\mathrm{tr}}F\|_{\mathrm{tr},n}^2=(1-o(1))\|\nabla_n^{\mathrm{tr}}F\|_{\mathrm{tr},n}^2.$$

The same estimate applied to differences makes $F\mapsto\nabla_n^{\mathrm{tr}}F$ closable from the form domain to the trace Hilbert space, so the estimate extends from smooth F to all $F\in\mathcal{D}(\mathcal{E}^{[n]})$. $\square$

A.2. **Discrete Palm adjoints.** For a smooth g on $\mathbb{T}^d$, set

$$\delta_l^{+,n}g(x_i^{[n]})=\frac{g(x_{i+e_l}^{[n]})-g(x_i^{[n]})}{h},\quad \delta_l^{-,n}g(x_i^{[n]})=\frac{g(x_i^{[n]})-g(x_{i-e_l}^{[n]})}{h},$$

$$(\nabla_ng)_l(x_i^{[n]})=\frac{1}{2}\big(\delta_l^{+,n}g(x_i^{[n]})+\delta_l^{-,n}g(x_i^{[n]})\big),\qquad \Delta_ng(x_i^{[n]})=\sum_{l=1}^d\frac{\delta_l^{+,n}g(x_i^{[n]})-\delta_l^{-,n}g(x_i^{[n]})}{h}.$$

If $\Gamma(\mu)=\Psi(\mu(f_1),\dots,\mu(f_m))$, define

$$(D_n\Gamma)_l(\iota^{[n]}s,x_i^{[n]})=\sum_{r=1}^m\partial_r\Psi\big(\iota^{[n]}s(f_1),\dots,\iota^{[n]}s(f_m)\big)(\nabla_nf_r)_l(x_i^{[n]}).$$

Proof of Theorem 5.7. By linearity it is enough to consider $W(\mu,x)=\Gamma(\mu)\nabla\varphi(x)$. For smooth F, the definition of the trace averages and the edge Palm measure give the exact identity

$$\langle\nabla_n^{\mathrm{tr}}F,(rW)_n\rangle_{\mathrm{tr},n}=-\int_{\Sigma^{[n]}}F(s)B_n^W(s)\,\mathcal{D}^{[n]}(ds), \tag{A.4}$$

where

$$B_n^W(s) = \sum_{i\in\mathcal{I}^{[n]}} \sum_{l=1}^{d} \frac{s_i - s_{i+e_l}}{h} \Gamma\Big(\iota^{[n]}(M_{i,l}s)\Big)\, \partial_l \varphi(x_{i,l}^{[n],\mathrm{mid}}). \tag{A.5}$$

Here $M_{i,l}s$ replaces the two masses (s_i, s_{i+e_l}) by their average. Reindexing the incoming part of (A.5) and Taylor expanding the cylinder coefficient around $\iota^{[n]}s$ gives

$$B_n^W(s) = \sum_i s_i \Gamma(\iota^{[n]}s)\Delta_n \varphi(x_i^{[n]}) + \sum_i s_i^2 D_n \Gamma(\iota^{[n]}s, x_i^{[n]}) \cdot \nabla_n \varphi(x_i^{[n]}) + \rho_n^W(s), \tag{A.6}$$

with $\|\rho_n^W\|_{H^{[n]}} \to 0$. The only terms that have to be discarded are adjacent mixed moments such as $\sum_{i,l} s_i s_{i+e_l} A_{i,l}(s)$ and $\sum_{i,l} s_i^2 s_{i+e_l} A_{i,l}(s)$ with uniformly bounded coefficients. Their $L^2(\mathcal{D}^{[n]})$ norms vanish, because a Dirichlet moment involving r distinct cells is $O(\alpha_n^r)$, while the number of adjacent edge pairs is only polynomial of order n^d.

The strong limit of (A.6) follows from the atomic moment consistency

$$J_n\left(\sum_i s_i^q A_n(\iota^{[n]}s, x_i^{[n]})\right) \longrightarrow \int_{\mathbb{T}^d} p(\mu,x)^{q-1} A(\mu,x)\,\mu(dx) \tag{A.7}$$

for $q = 1, 2$, whenever $A_n \to A$ uniformly and the A_n are uniformly bounded cylinder quantities. This is proved by writing $\mu = \sum_a p_a \delta_{Y_a}$: the diagonal contributions converge to $\sum_a p_a^q A(\mu, Y_a)$, while off-diagonal atoms eventually lie in different cells. Applying (A.7) to the two leading terms in (A.6) gives

$$J_n B_n^W \longrightarrow \int_{\mathbb{T}^d} [\Gamma(\mu)\Delta\varphi(x) + p(\mu,x) D\Gamma(\mu,x)\cdot\nabla\varphi(x)]\,\mu(dx) = B_W(\mu).$$

For the canonical endpoint weights above, the remainder in (5.115) can be taken to be $R_n^W = 0$. If smoothed endpoint cutoffs are used instead, the same computation produces an error R_n^W with norm tending to zero.

It remains to identify the trace-test norm. Expanding the definition gives

$$\|(rW)_n\|_{\mathrm{tr},n}^2 = \frac{1}{2}\sum_{i,l} \int_{\Sigma^{[n]}} (s_i + s_{i+e_l})^3 \left| W_l\Big(\iota^{[n]}(s^{e,1/2}), x_{i,l}^{[n],\mathrm{mid}}\Big)\right|^2 \mathcal{D}^{[n]}(ds).$$

The mixed cubic terms vanish by the same adjacent-moment estimate, and the diagonal cubic terms converge by (A.7) with $q = 3$. This yields (5.117). □

A.3. **Continuum Palm duality.**

Proof of Theorem 5.8. Let

$$\mathsf{H}_T = L^2\Big(P(\mathbb{T}^d) \times \mathbb{T}^d, \mathcal{D}(d\mu)\mu(dx); \mathbb{R}^d\Big)$$

and let $\mathsf{D} : \mathcal{D}(\mathcal{E}) \subset H \to \mathsf{H}_T$ be the closed Wasserstein gradient. For $W \in \mathscr{W}$, set $g_W(\mu,x) = p(\mu,x)W(\mu,x)$.

First, write $\pi(q,X) = \sum_k q_k \delta_{X_k}$ for the labelled atomic projection. For $U \in \mathcal{A}$ and $W = \Gamma\nabla\varphi$, the labelled atomic representation $\mu = \sum_k q_k \delta_{X_k}$ gives, by integration by parts in each torus variable X_k,

$$\int DU(\mu,x)\cdot g_W(\mu,x)\,\mu(dx)\mathcal{D}(d\mu) = -\int U(\mu) B_W(\mu)\,\mathcal{D}(d\mu). \tag{A.8}$$

The extra factor $p(\mu,x)$ in g_W is exactly the factor produced when differentiating a cylinder coefficient with respect to a tagged atom location. Since $B_W \in L^\infty(\mathcal{D})$ and $g_W \in \mathsf{H}_T$, (A.8) extends by closure to every $U \in \mathcal{D}(\mathcal{E})$. Equivalently,

$$g_W \in \mathcal{D}(\mathsf{D}^*), \qquad \mathsf{D}^* g_W = -B_W. \tag{A.9}$$

Second, the graph set $\{(g_W, \mathsf{D}^* g_W) : W \in \mathscr{W}\}$ is dense in $\mathrm{Graph}(\mathsf{D}^*)$. The proof is by the standard adjoint graph-density criterion. If $(Y,U) \in \mathsf{H}_T \times H$ annihilates all these graph pairs, then

$$\int U B_W \, d\mathcal{D} = \int Y(\mu,x)\cdot p(\mu,x) W(\mu,x)\,\mu(dx)\mathcal{D}(d\mu) \qquad \forall W \in \mathscr{W}.$$

In labelled variables this says that the distributional derivatives of $U \circ \pi$ are $-q_k Y(\pi(q,X), X_k)$ in the variables X_k. The density of the first-jet class generated by the fields pW then implies $U \in \mathcal{D}(\mathcal{E})$ and $DU = -Y$. The density statement is obtained by localizing to finitely many labelled atoms with separated masses and separated spatial locations. For $d \geq 2$, finite collision diagonals have zero $W^{1,2}$-capacity, so these localizations lose no graph norm. On each separated chart, one reconstructs the tagged atoms from the unlabelled measure and approximates the corresponding labelled Sobolev functions by ordinary smooth cylinders. Hence the annihilator of the Palm graph equals the annihilator of $\mathrm{Graph}(\mathsf{D}^*)$, proving graph density.

Now suppose $U \in \mathcal{D}(\mathcal{E})$. By (A.9),

$$\begin{aligned} -2\int U B_W \, d\mathcal{D} - \|g_W\|^2_{\mathsf{H}_T} &= 2\langle \mathsf{D}U, g_W\rangle_{\mathsf{H}_T} - \|g_W\|^2_{\mathsf{H}_T} \\ &= \|\mathsf{D}U\|^2_{\mathsf{H}_T} - \|\mathsf{D}U - g_W\|^2_{\mathsf{H}_T} \leq \mathcal{E}(U,U). \end{aligned}$$

Graph density lets us choose W_m with $g_{W_m} \to DU$ in H_T, giving equality after taking the supremum.

If $U \notin \mathcal{D}(\mathcal{E})$ and the supremum were finite, the same quadratic estimate applied to λW for all $\lambda \in \mathbb{R}$ would give

$$|\langle U, \mathsf{D}^* g_W\rangle_H| \leq C\|g_W\|_{\mathsf{H}_T} \qquad \forall W \in \mathscr{W}.$$

By graph density this extends to all $g \in \mathcal{D}(\mathsf{D}^*)$, so $U \in \mathcal{D}(\mathsf{D}^{**}) = \mathcal{D}(\mathsf{D}) = \mathcal{D}(\mathcal{E})$, a contradiction. Therefore the supremum is $+\infty$ outside the form domain. This proves (5.118). $\square$

References

[1] François Delarue, Mattia Martini, and Giacomo Enrico Sodini. Hjb equations driven by the dirichlet-ferguson laplacian in wasserstein-sobolev spaces, 2025. URL `https://arxiv.org/abs/2511.03522`.

[2] Lorenzo Dello Schiavo. The dirichlet–ferguson diffusion on the space of probability measures over a closed riemannian manifold. *Annals of Probability*, 50(2):591–648, 2022. doi: 10.1214/21-AOP1541. URL `https://doi.org/10.1214/21-AOP1541`.

[3] Massimo Fornasier, Giuseppe Savaré, and Giacomo Enrico Sodini. Density of subalgebras of Lipschitz functions in metric Sobolev spaces and applications to Wasserstein Sobolev spaces. *Journal of Functional Analysis*, 285(11):110153, 2023. doi: 10.1016/j.jfa.2023.110153.

[4] Masatoshi Fukushima. *Dirichlet Forms and Markov Processes*, volume 23 of *North-Holland Mathematical Library*. North-Holland Publishing Co., Amsterdam, 1980. ISBN 978-0-444-85421-6.

[5] Masatoshi Fukushima, Yoichi Oshima, and Masayoshi Takeda. *Dirichlet Forms and Symmetric Markov Processes*, volume 19 of *de Gruyter Studies in Mathematics*. de Gruyter, Berlin, 2 edition, 2010.

[6] Nicola Gigli. *Second Order Analysis on* $(\mathscr{P}_2(M), W_2)$, volume 216 of *Memoirs of the American Mathematical Society*. American Mathematical Society, Providence, RI, 2012. ISBN 978-0-8218-8529-1.

[7] Olav Kallenberg. *Foundations of Modern Probability*, volume 99 of *Probability Theory and Stochastic Modelling*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

[8] Wei Liu and Michael Röckner. *Stochastic Partial Differential Equations: An Introduction*. Universitext. Springer, Cham, 2015. ISBN 978-3-319-22353-7. doi: 10.1007/978-3-319-22354-4.

[9] William McLean. *Strongly Elliptic Systems and Boundary Integral Equations*. Cambridge University Press, Cambridge, 2000. ISBN 0-521-66332-6.

[10] Felix Otto. The geometry of dissipative evolution equations: the porous medium equation. *Communications in Partial Differential Equations*, 26(1-2):101–174, 2001. doi: 10.1081/PDE-100002243.

[11] Daniel Revuz and Marc Yor. *Continuous Martingales and Brownian Motion*, volume 293 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, Heidelberg, 3 edition, 1999. ISBN 978-3-540-64325-8. doi: 10.1007/978-3-662-06400-9.

Department, Institution, City, Country
Email address: `you@example.com`